Vertex Covers

Alexander Golovnev

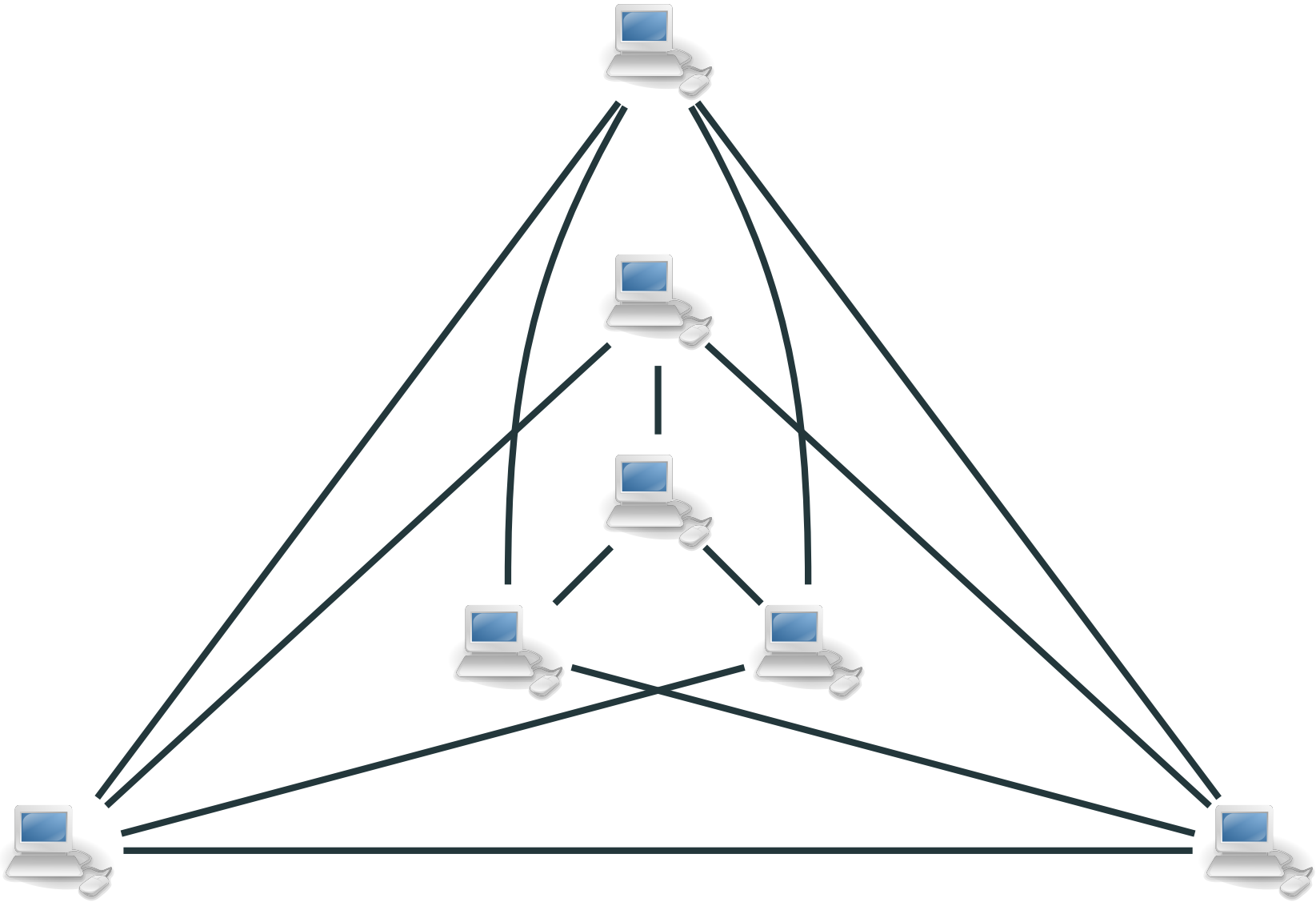
Outline

Antivirus System

Vertex Covers

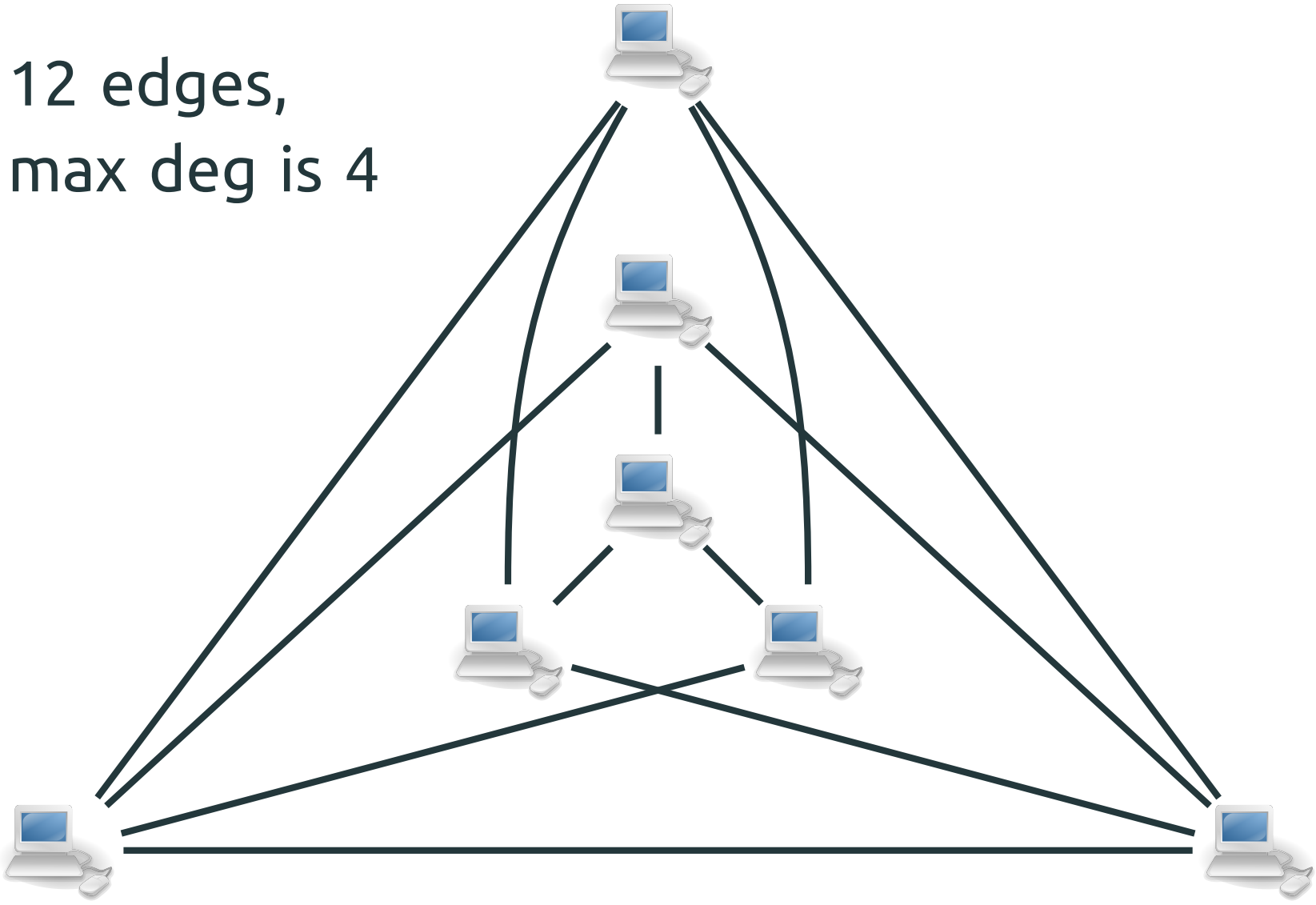
König's Theorem

Antivirus System



Antivirus System

12 edges,
max deg is 4



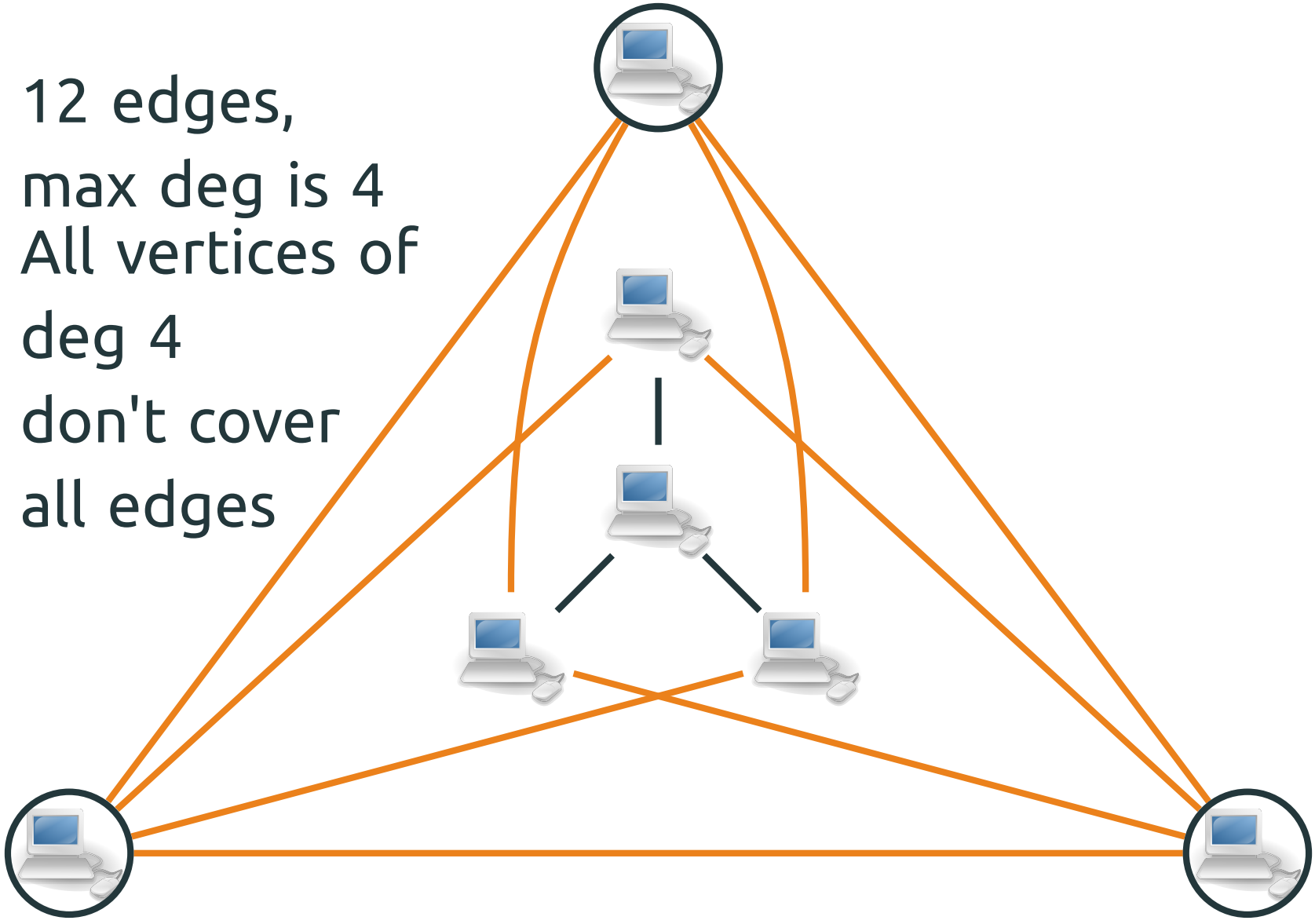
Antivirus System

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All vertices of deg 4

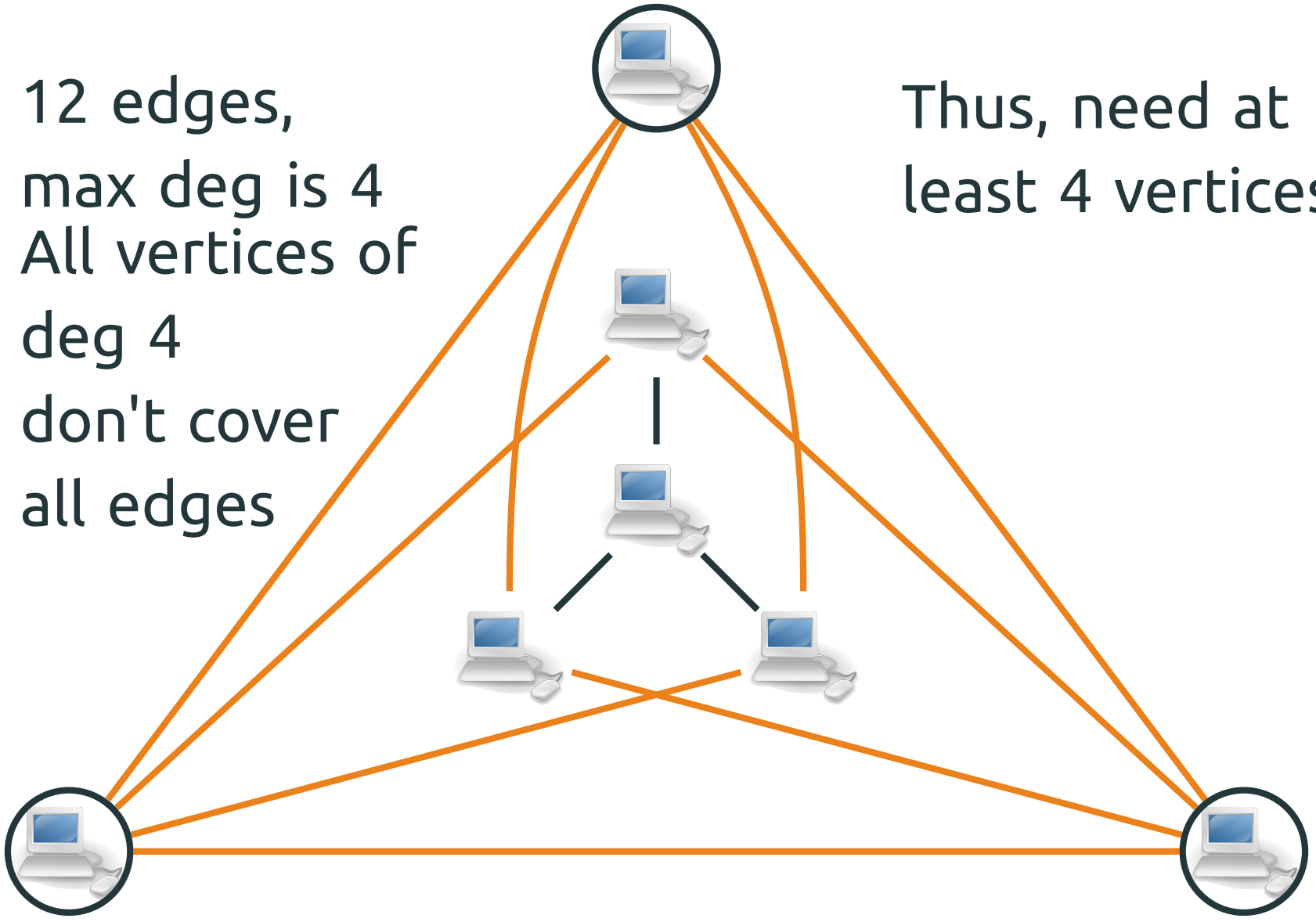
don't cover all edges



Antivirus System

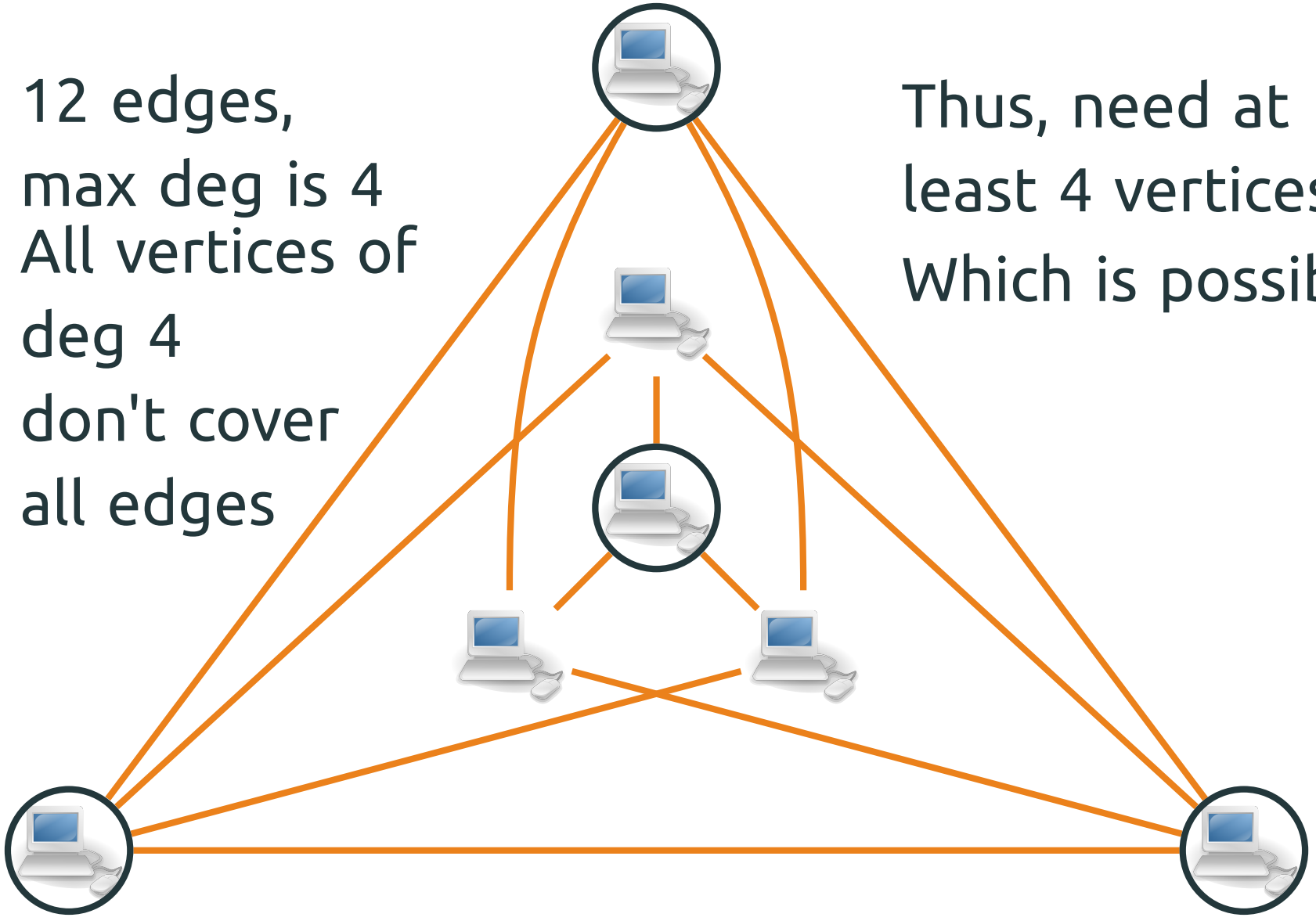
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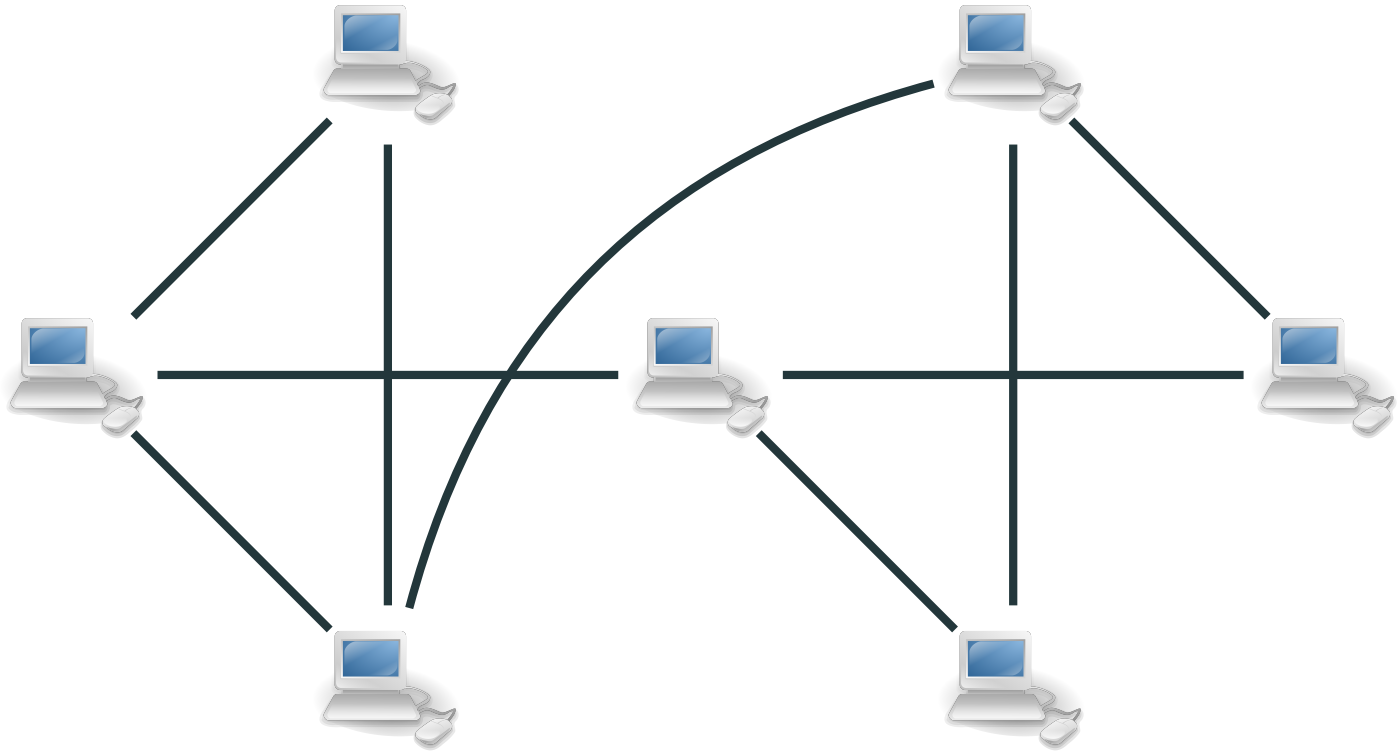
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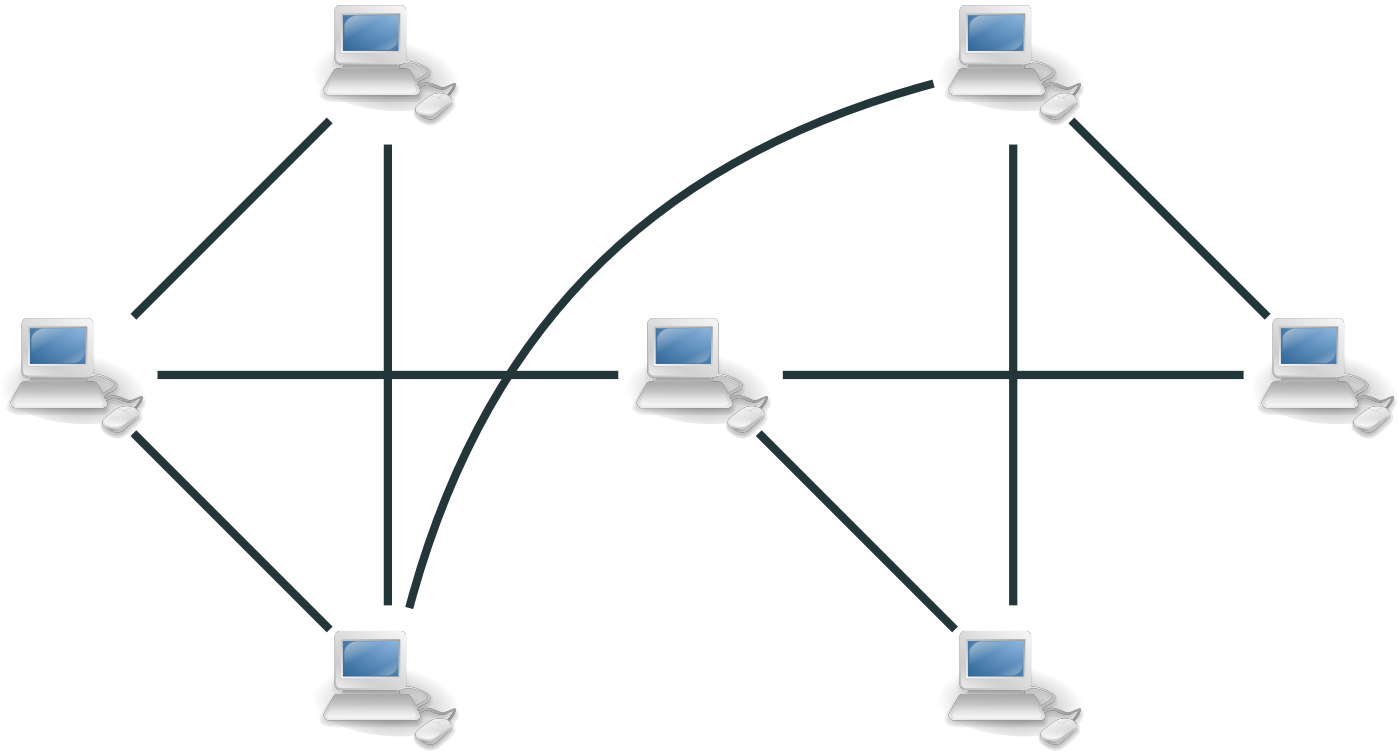
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Which is possible

Antivirus System



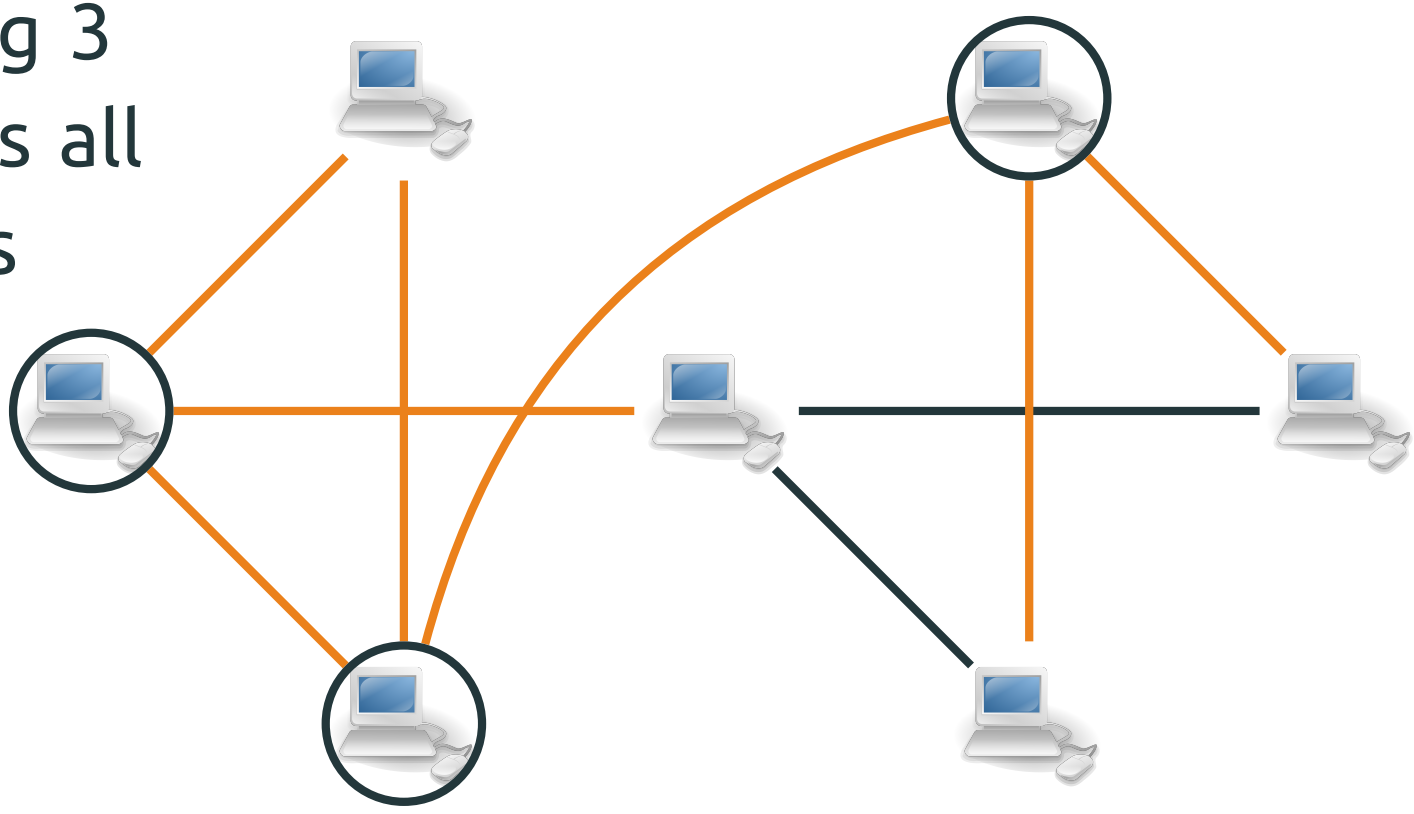
Antivirus System

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Antivirus System

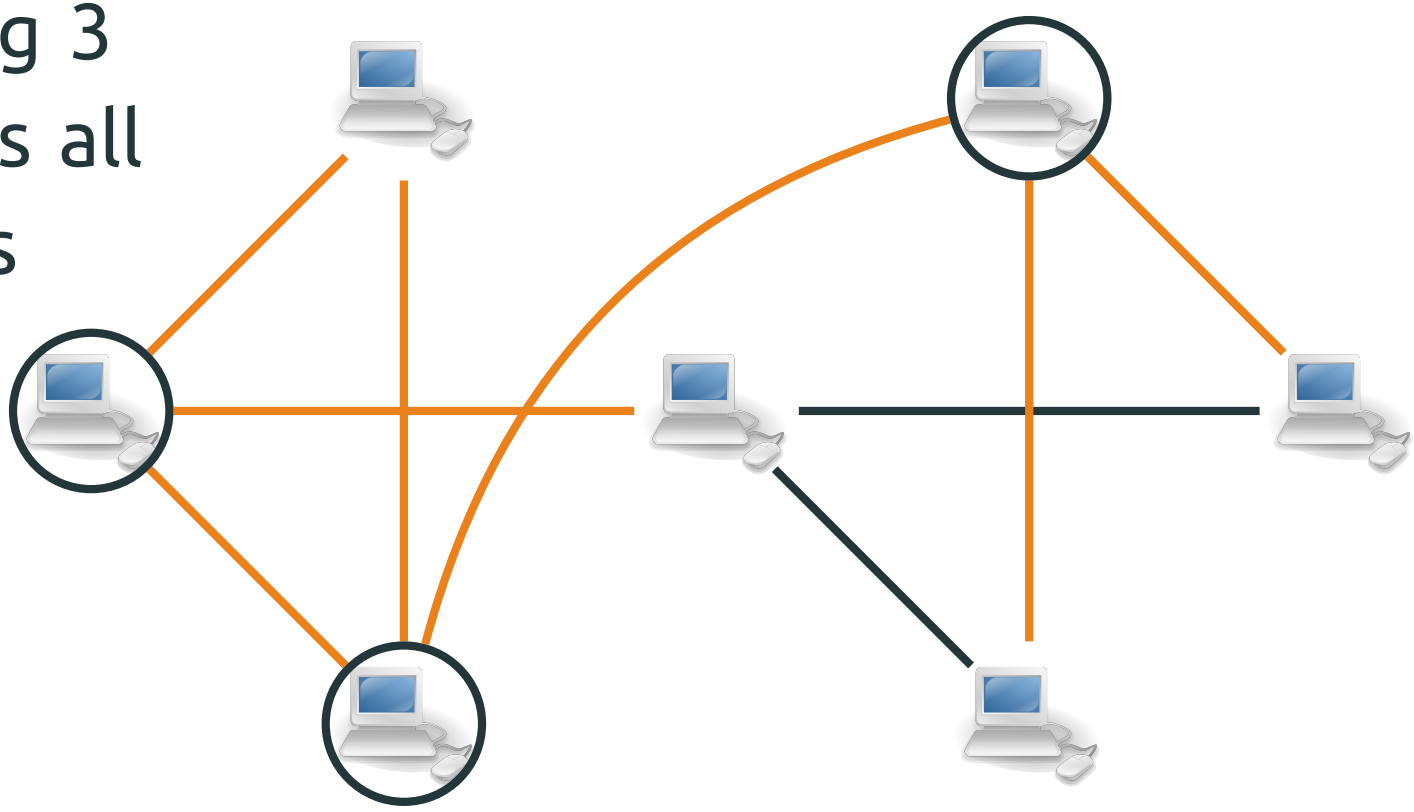
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Antivirus System

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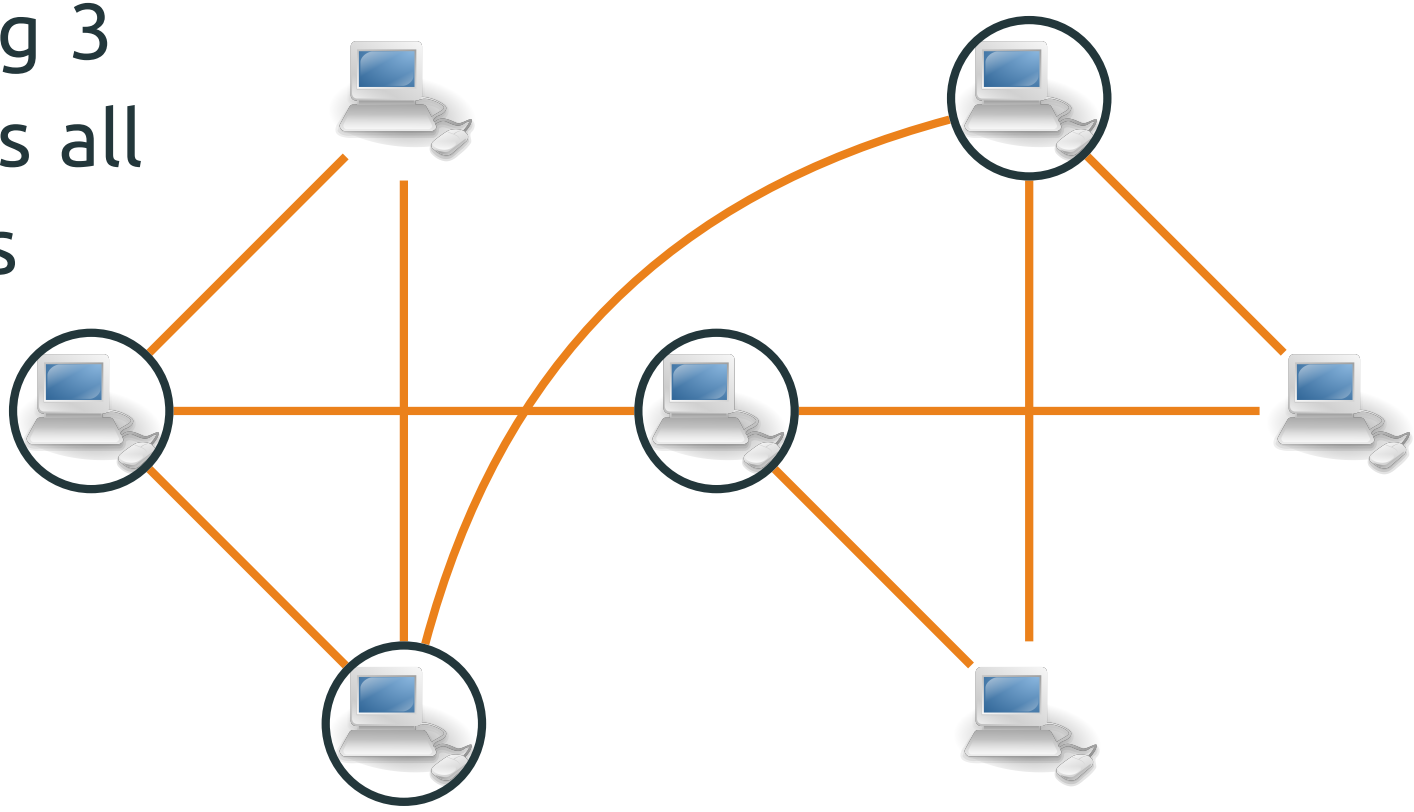
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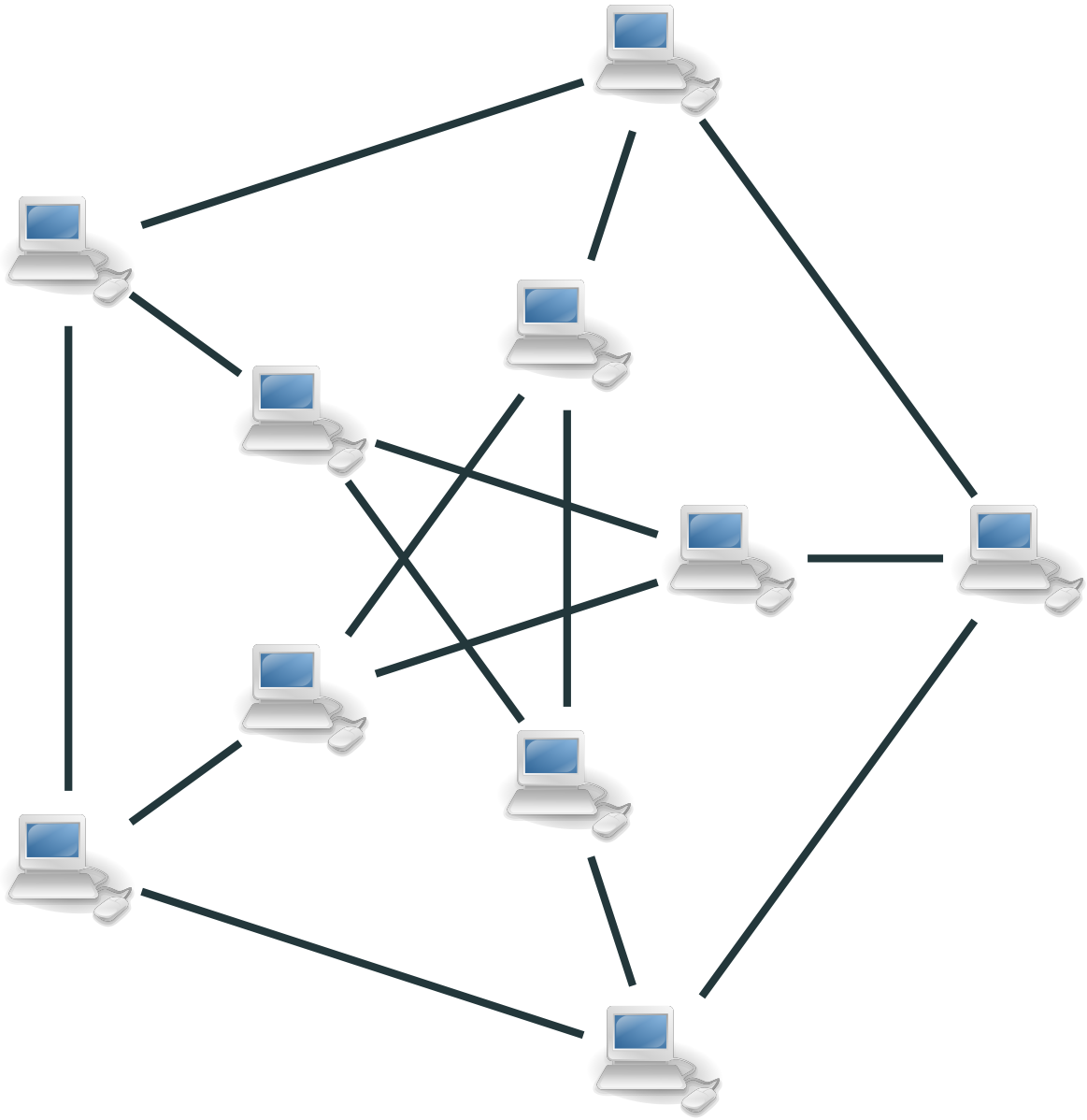
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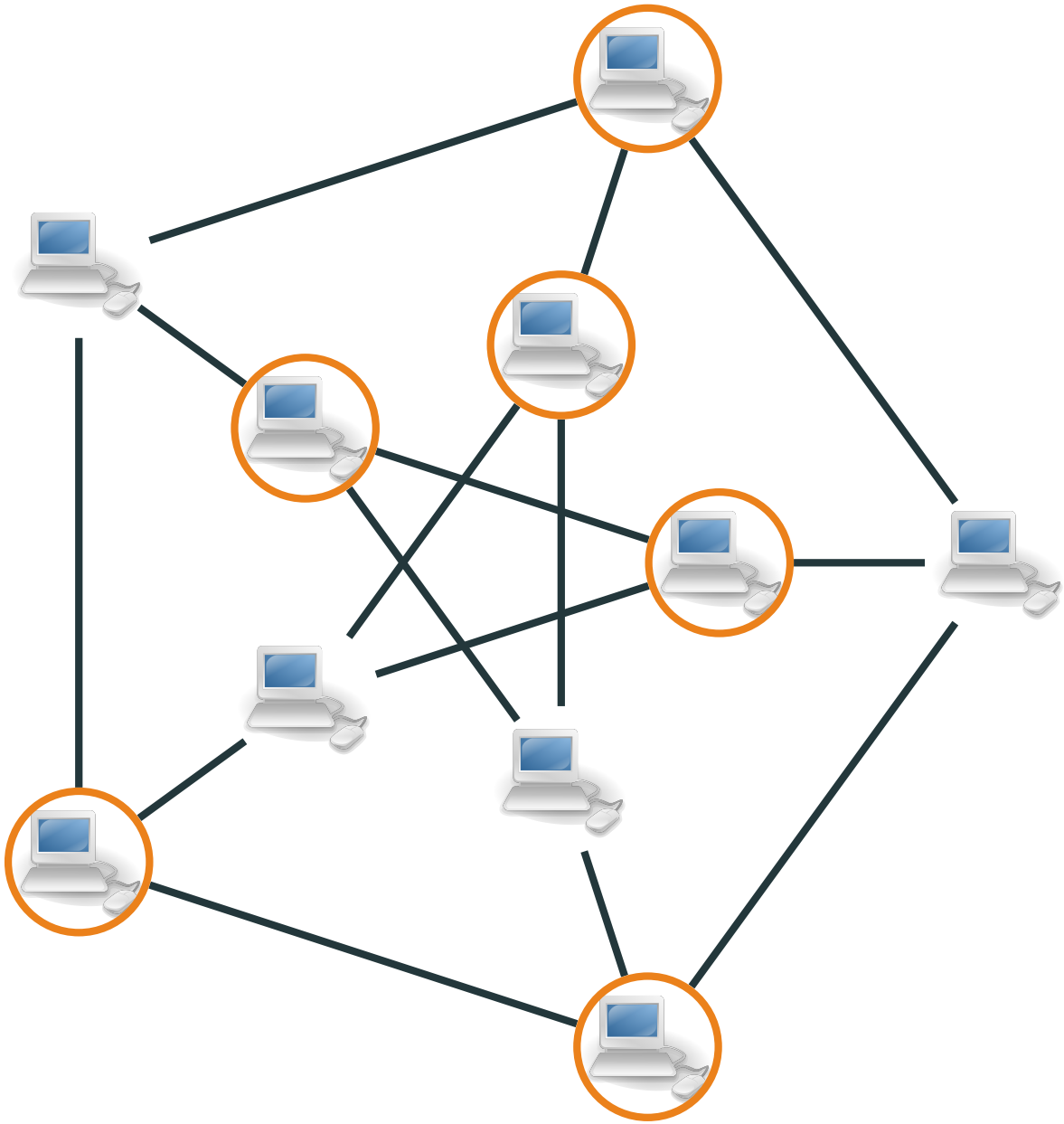
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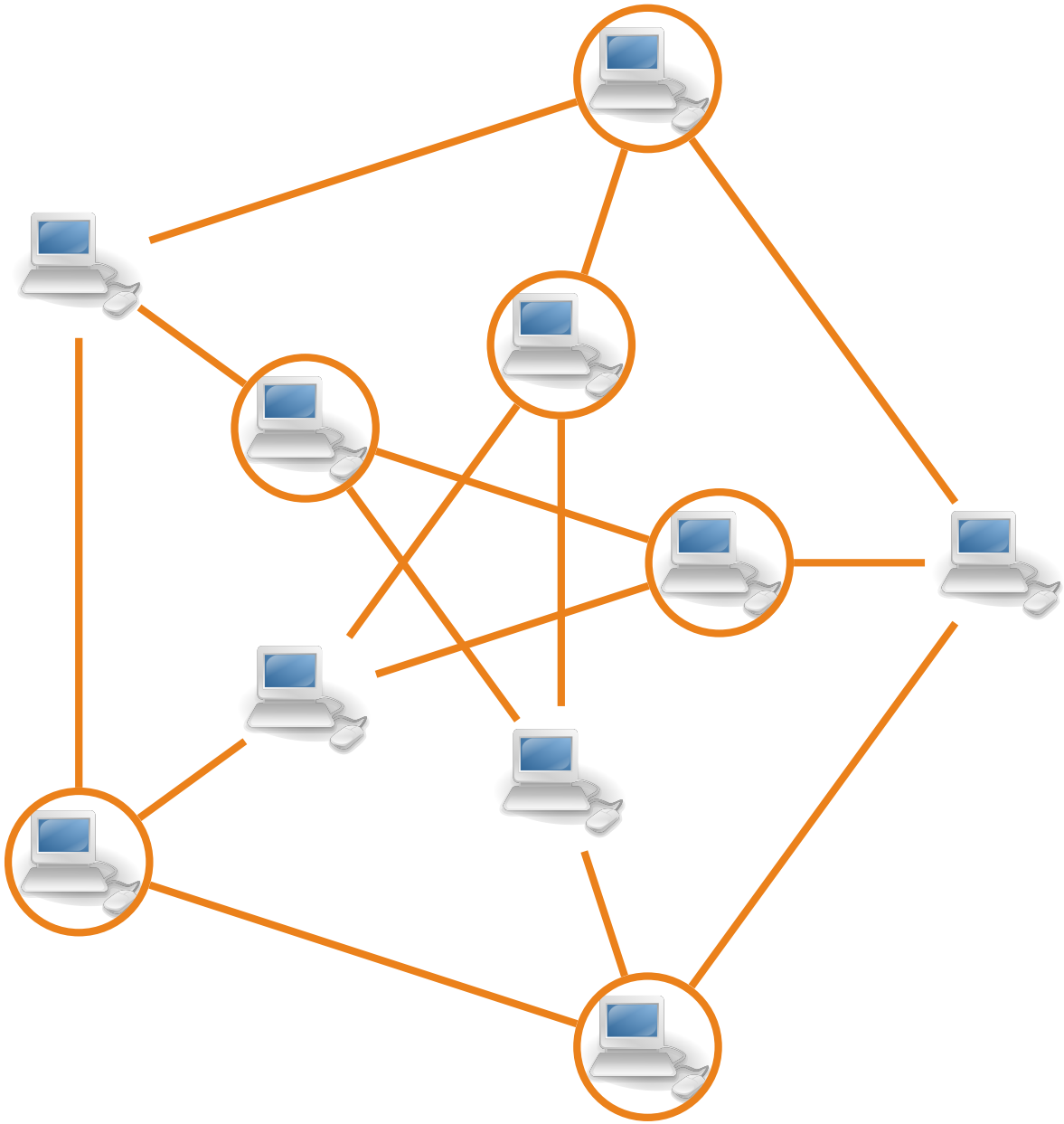
Antivirus System



Antivirus System



Antivirus System



Outline

Antivirus System

Vertex Covers

König's Theorem

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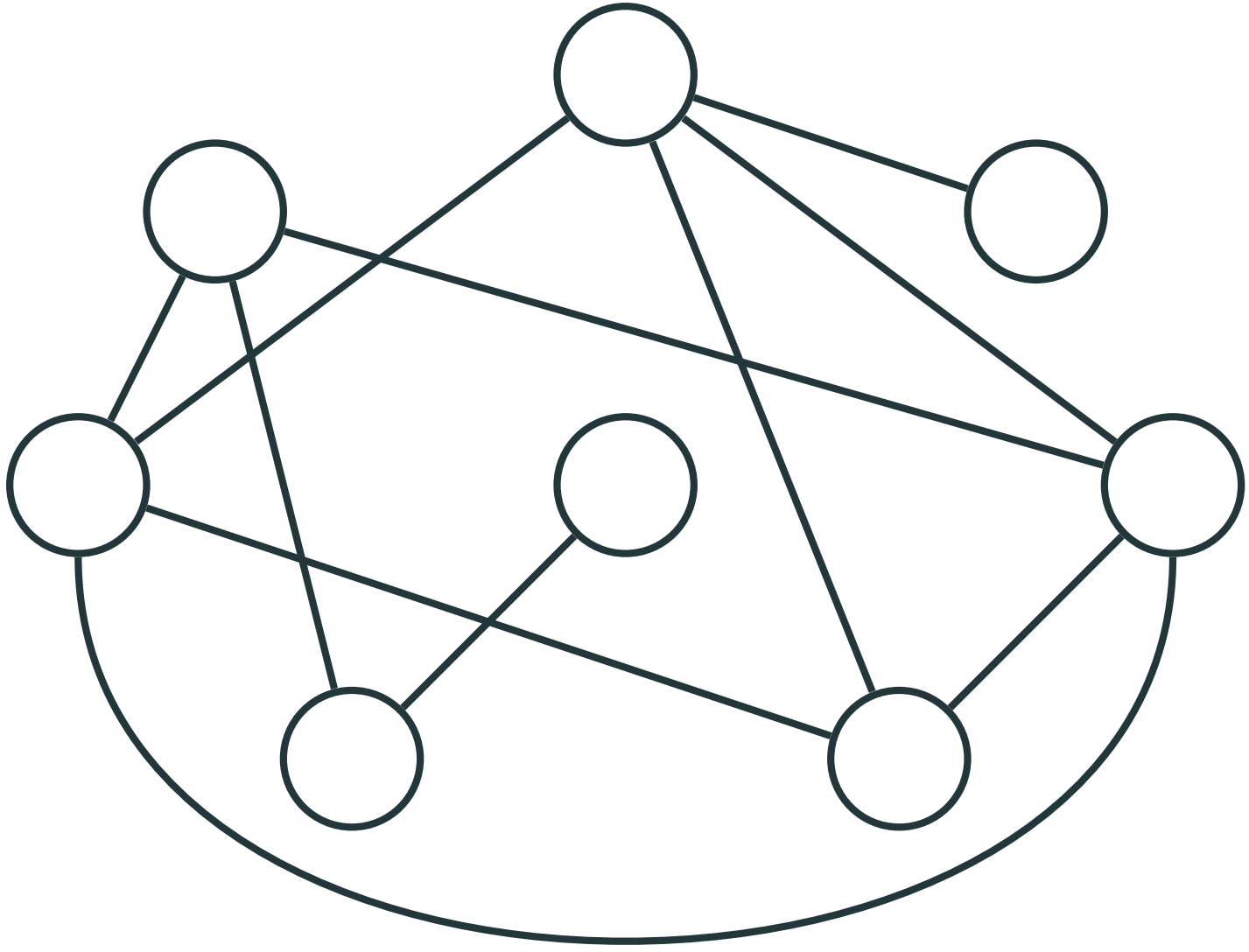
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Vertex Covers

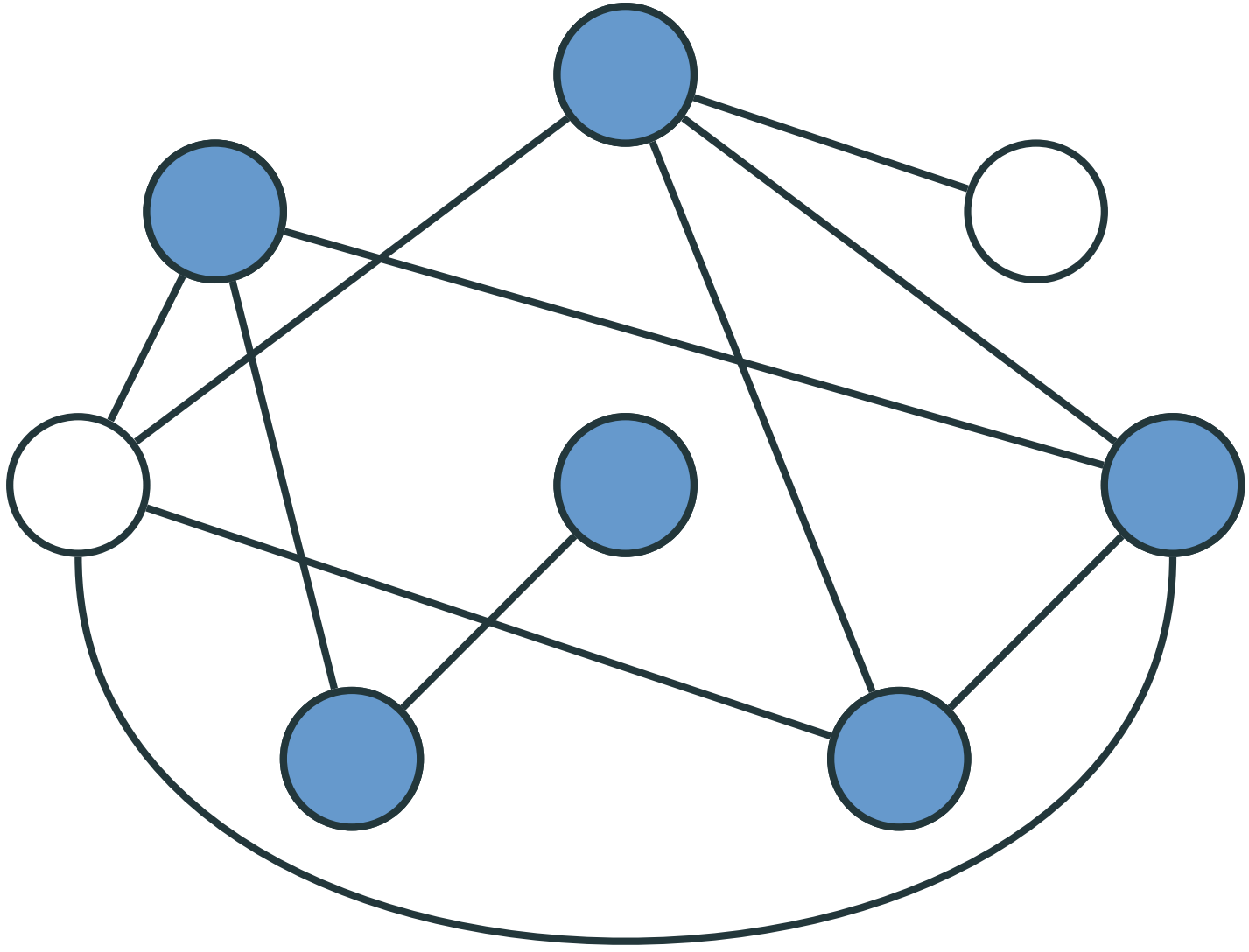
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- The Size of a Minimum Vertex Cover is denoted by $\beta(G)$.

Vertex Covers: Examples



Vertex Covers: Examples

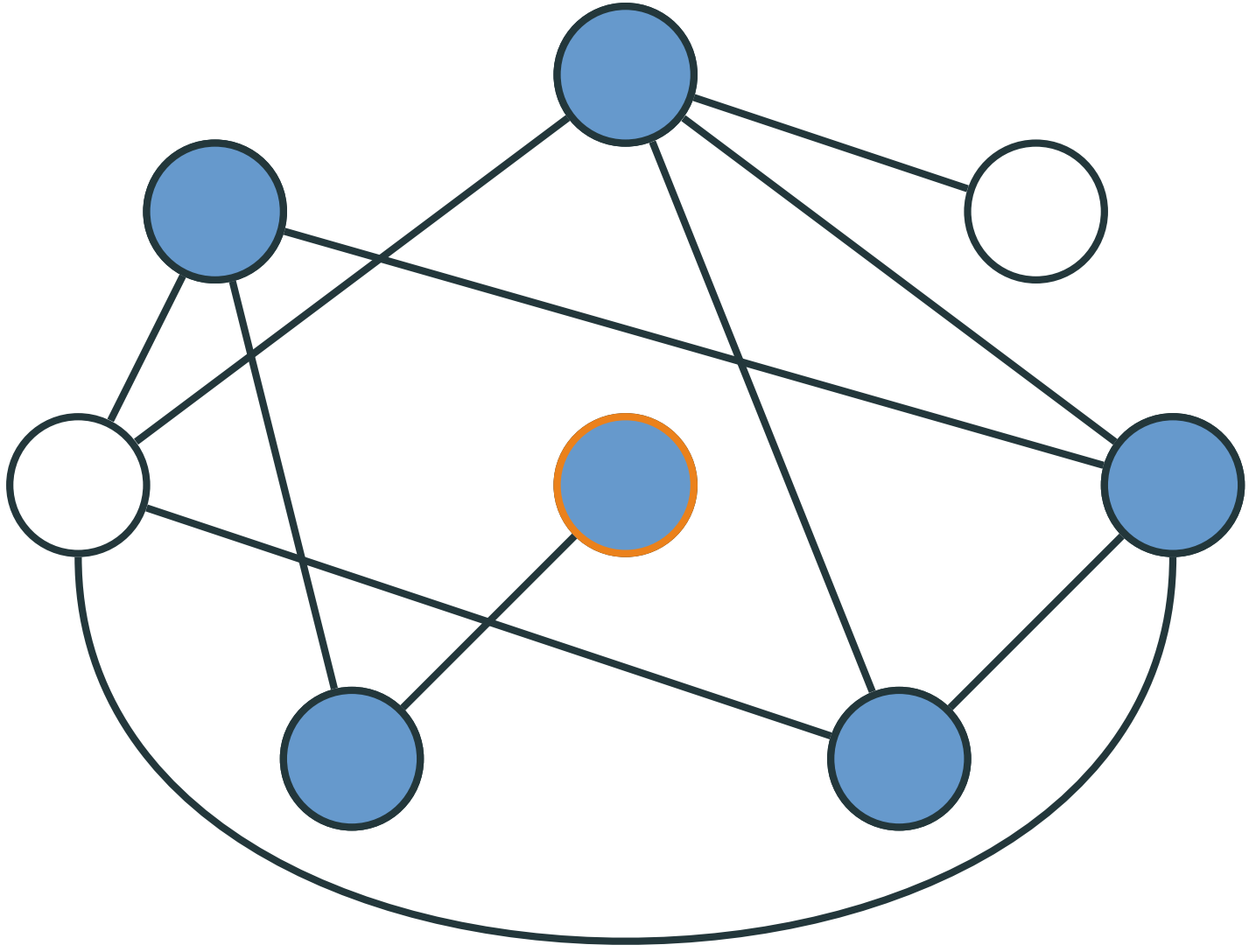
A Vertex Cover



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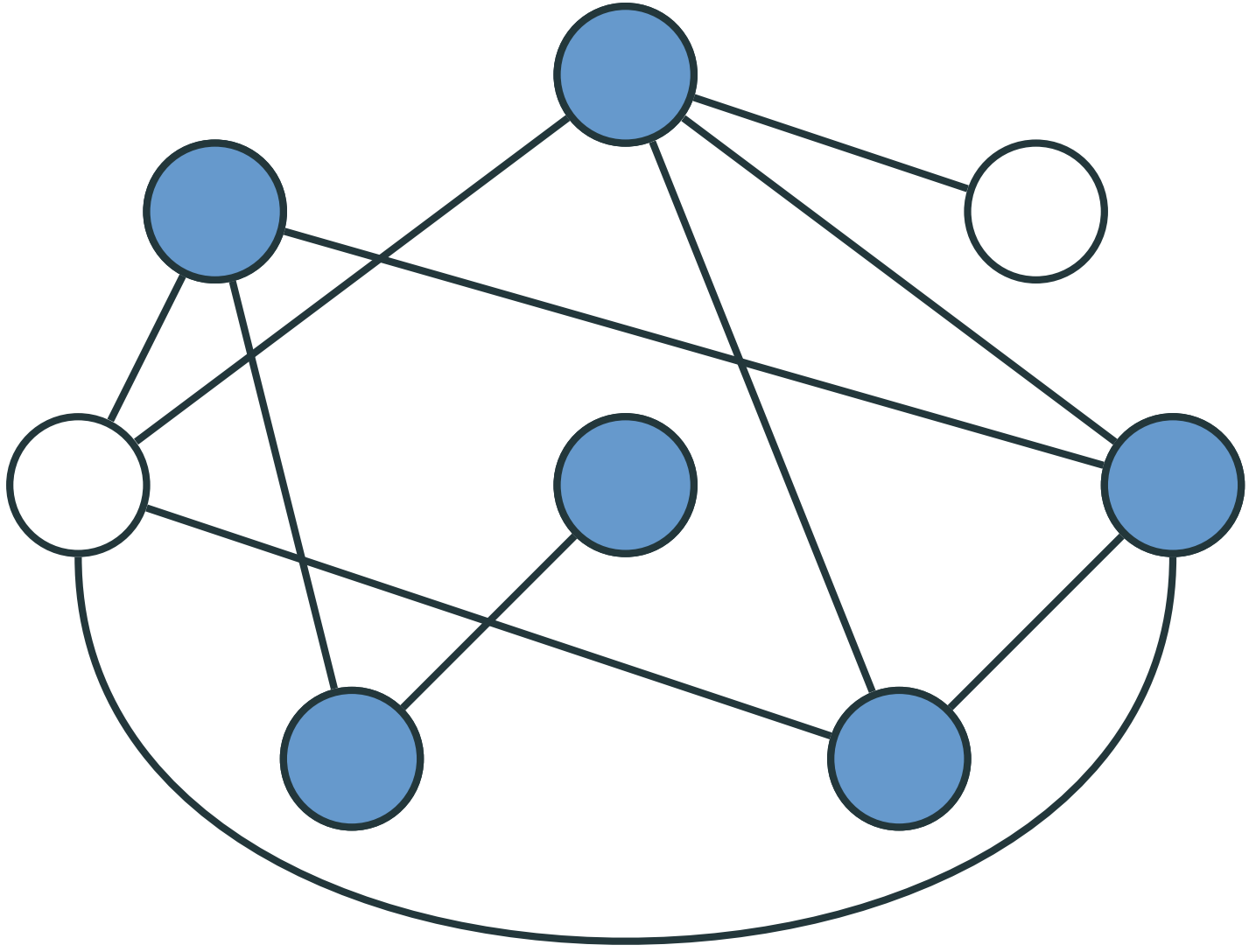
A Vertex Cover

Not a Mini-mal VC

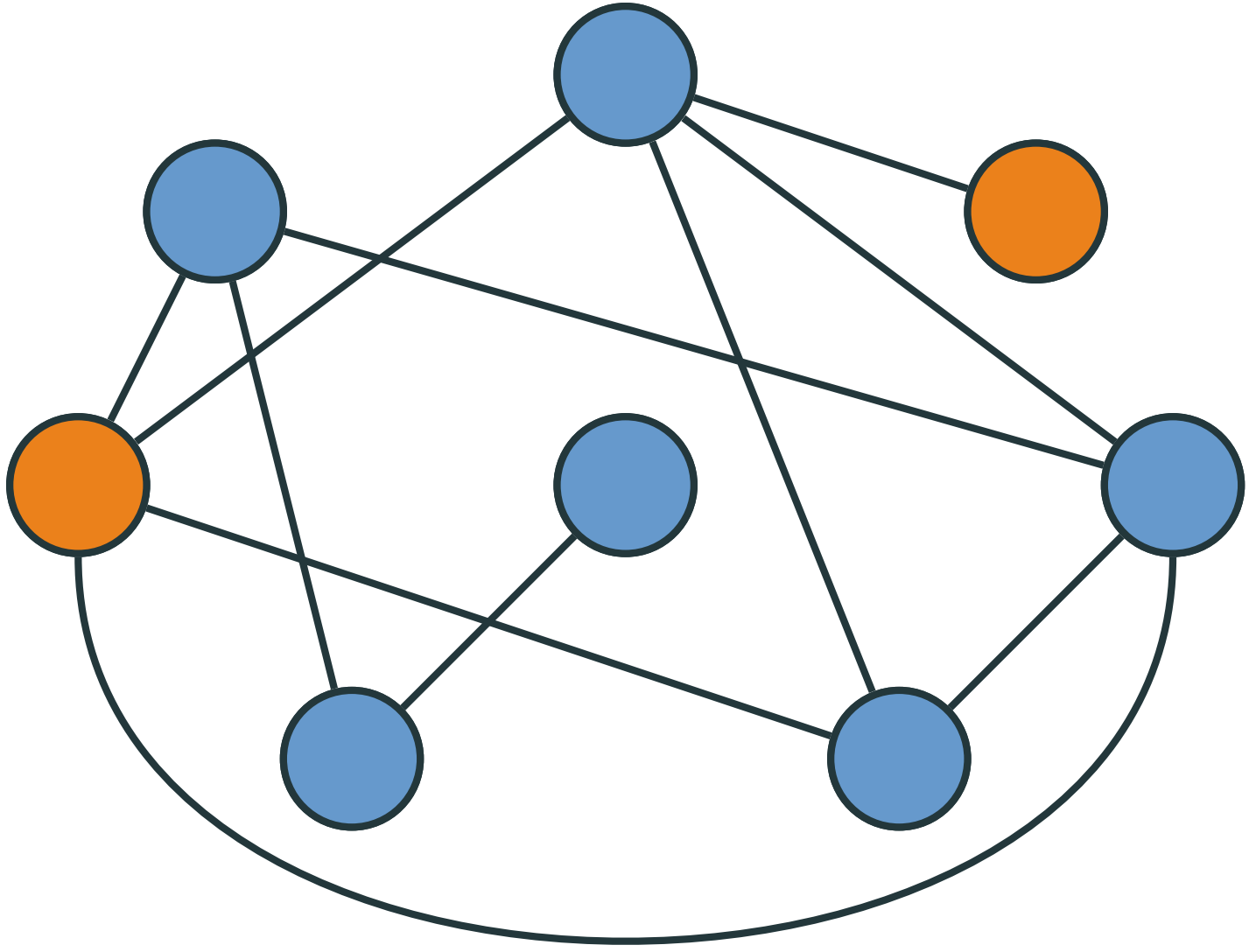


Vertex Covers: Examples

A Vertex Cover

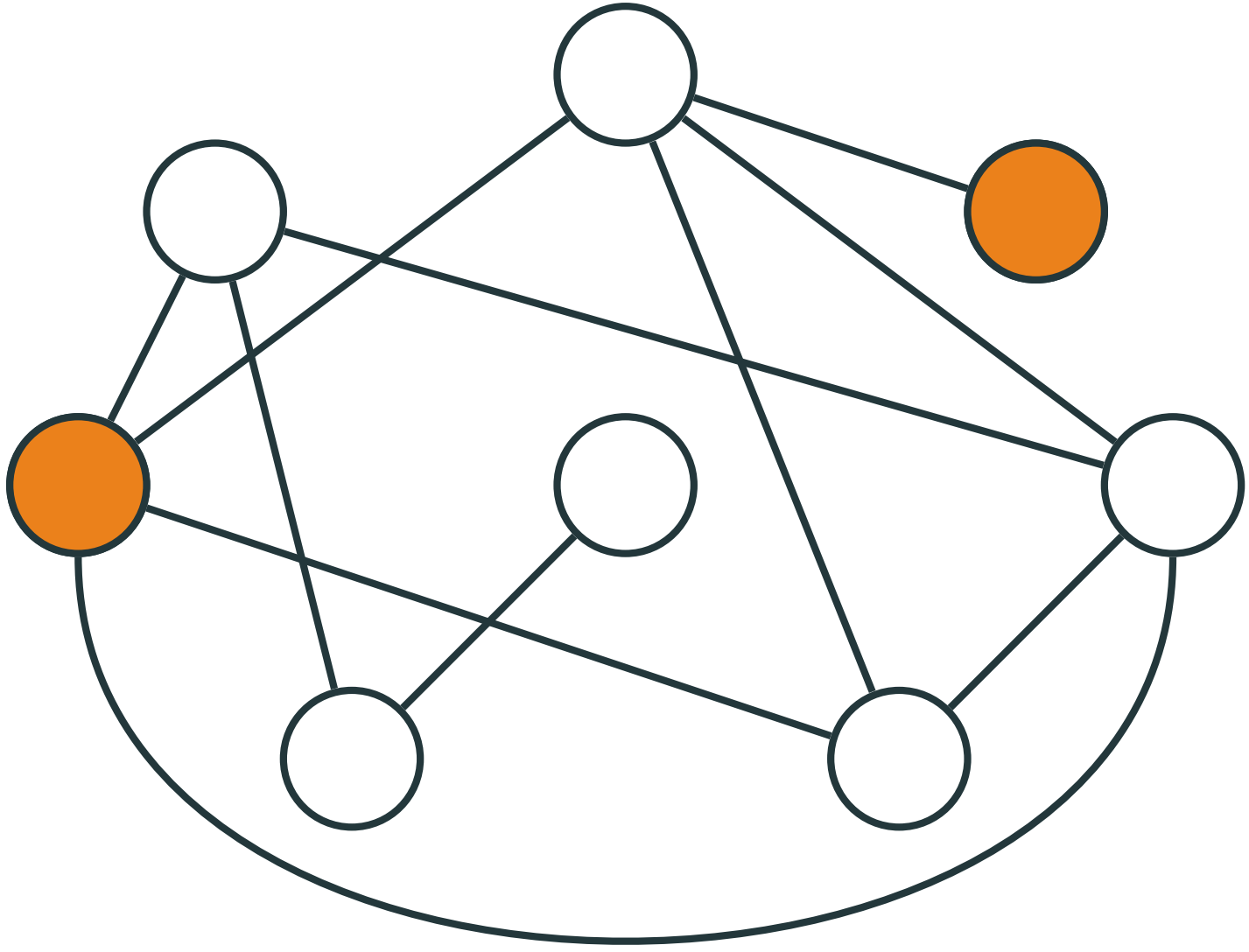


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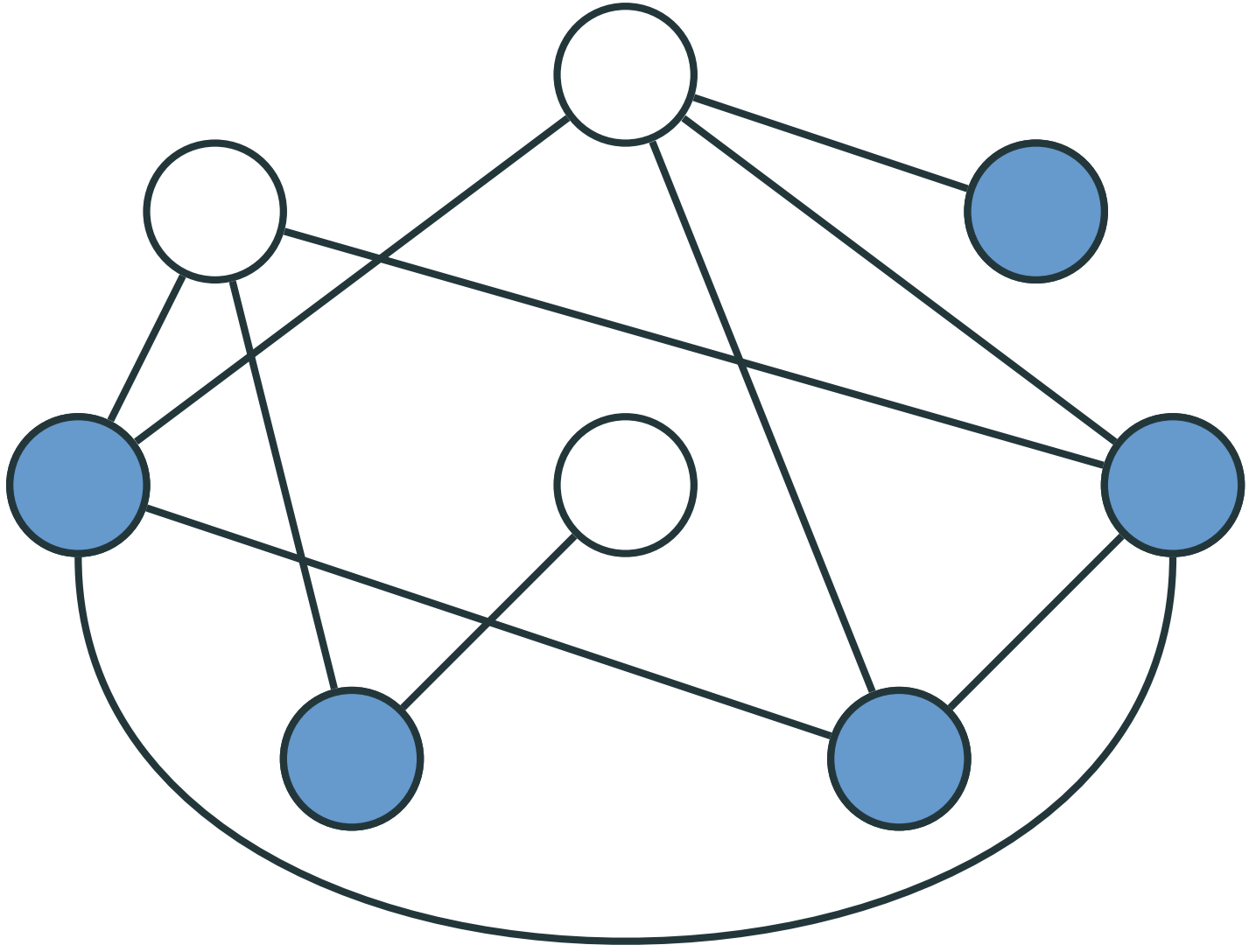
Vertex Covers: Examples

An Independent Set



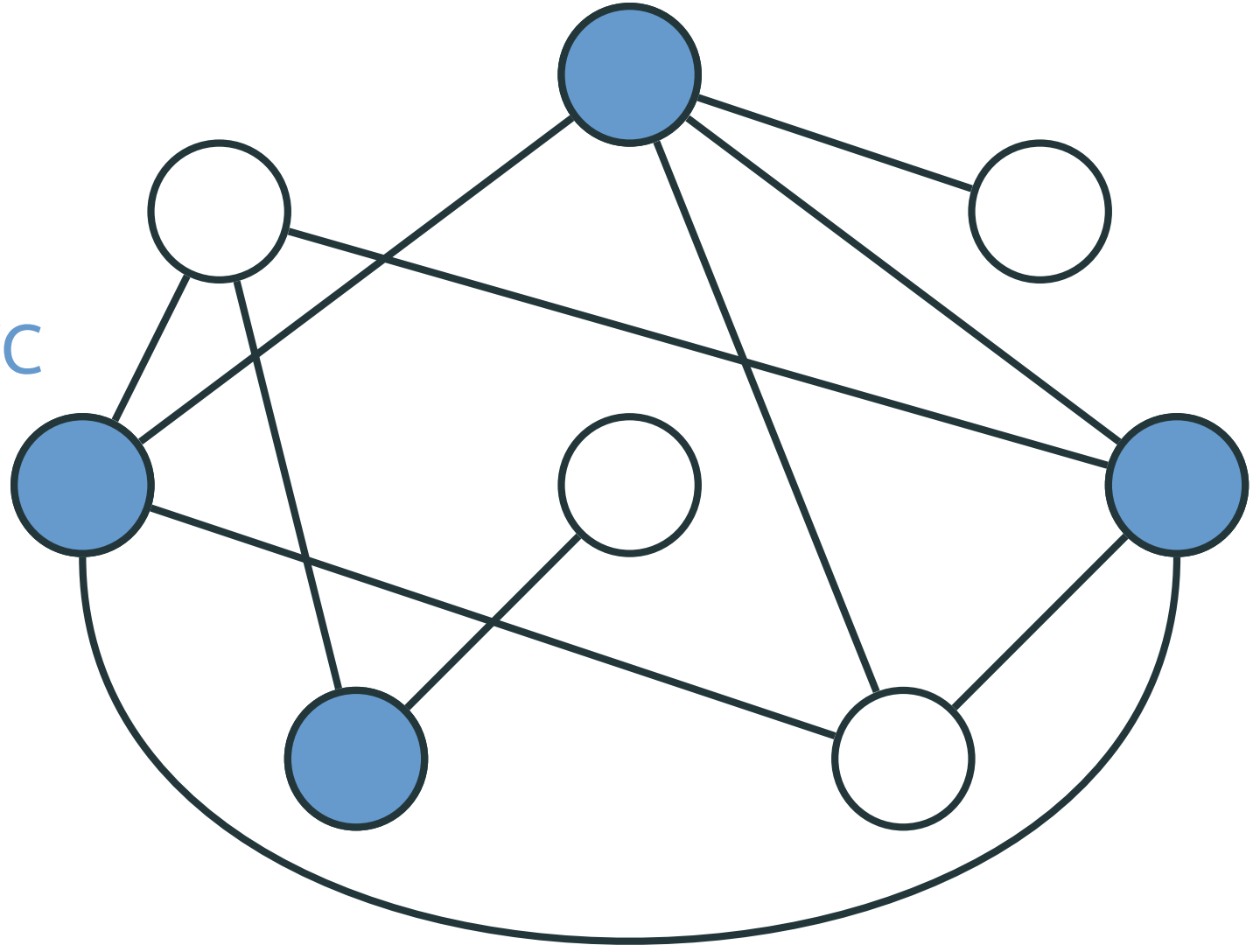
Vertex Covers: Examples

A Minimal VC



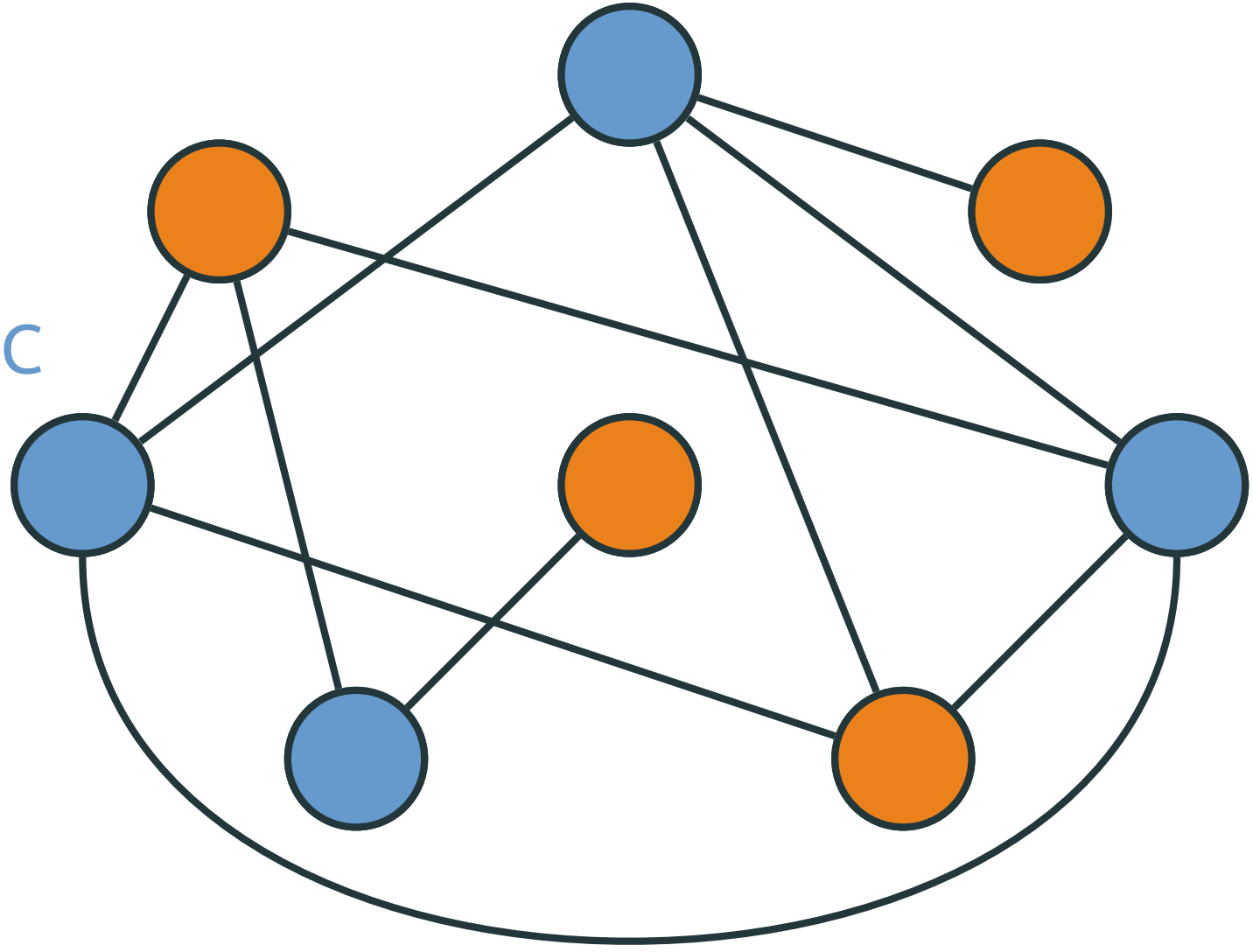
Vertex Covers: Examples

A Minimum VC



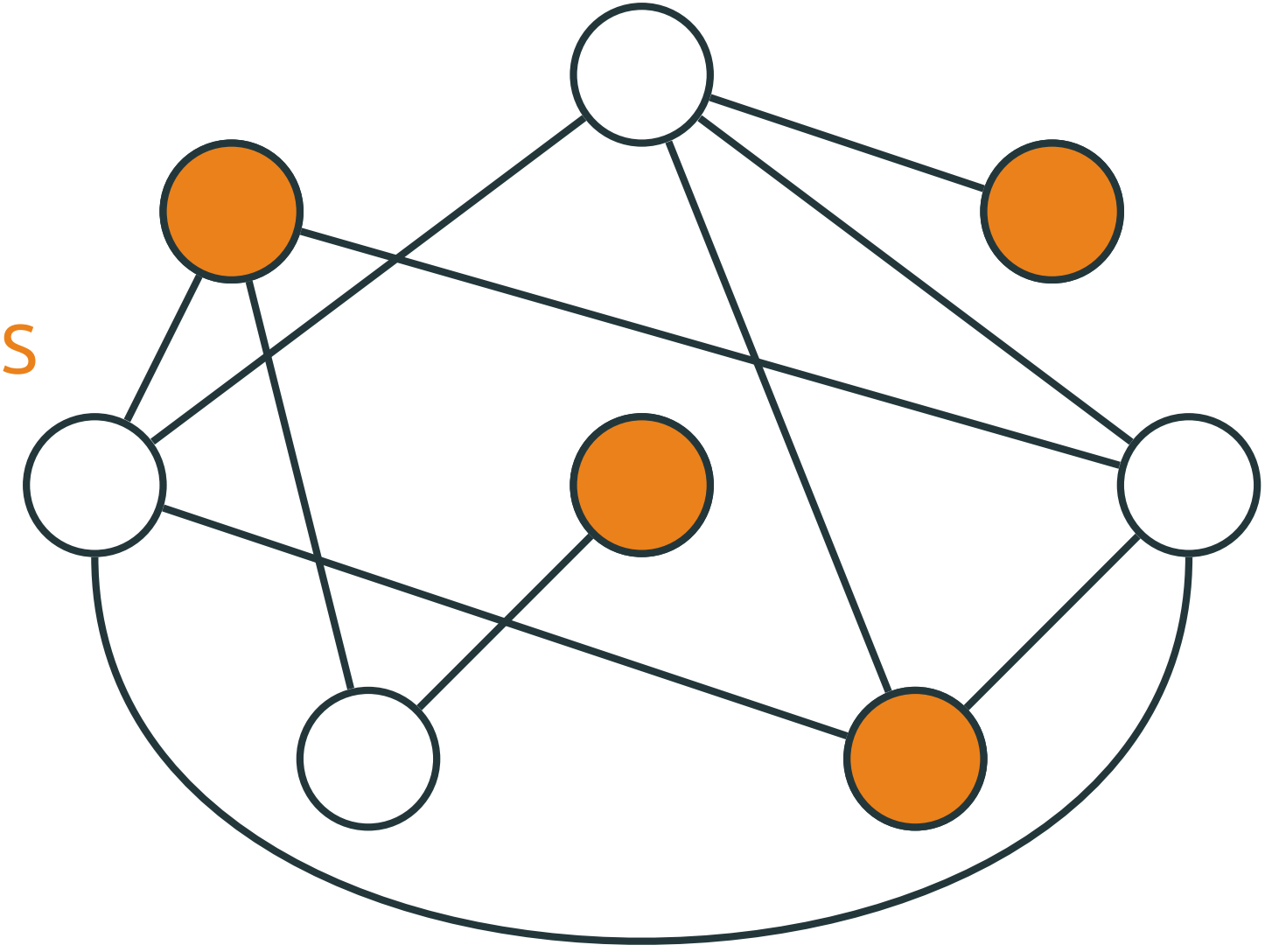
Vertex Covers: Examples

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Vertex Covers: Examples

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Vertex Covers and IS's

Fact

A set of vertices is a Vertex Cover if and only if its complement is an Independent Set

Vertex Covers and IS's

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Corollary

For every graph G on n vertices:

$$\beta(G) + \alpha(G) = n \, .$$

Outline

Antivirus System

Vertex Covers

König's Theorem

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Fact

The vertices of any Maximal Matching form a (not necessarily minimal) Vertex Cover.

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Theorem (König, 1931)

In a bipartite graph, the number of edges in a Maximum Matching equals the number of vertices in a Minimum Vertex Cover.

König's Theorem. Proof

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In a bipartite graph $G = ((L \cup R), E)$, the number of edges in a Maximum Matching equals the number of vertices in a Minimum Vertex Cover.

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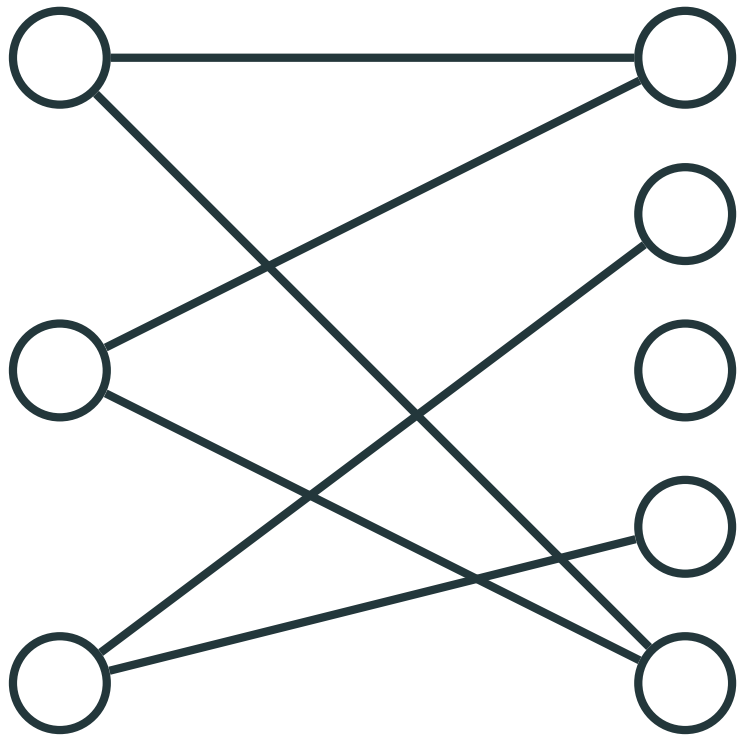
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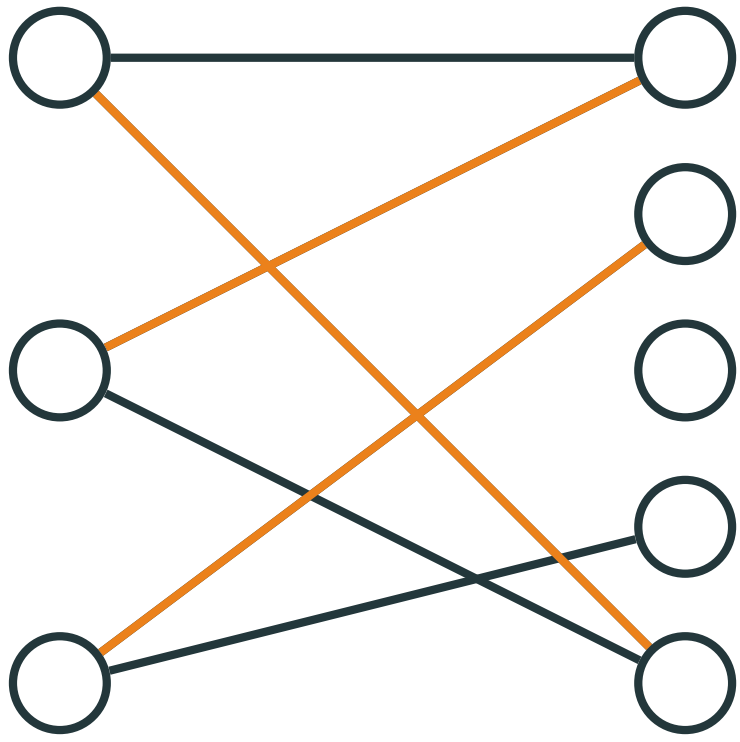
Proof:

- A VC must contain at least 1 vertex from each edge of a Matching

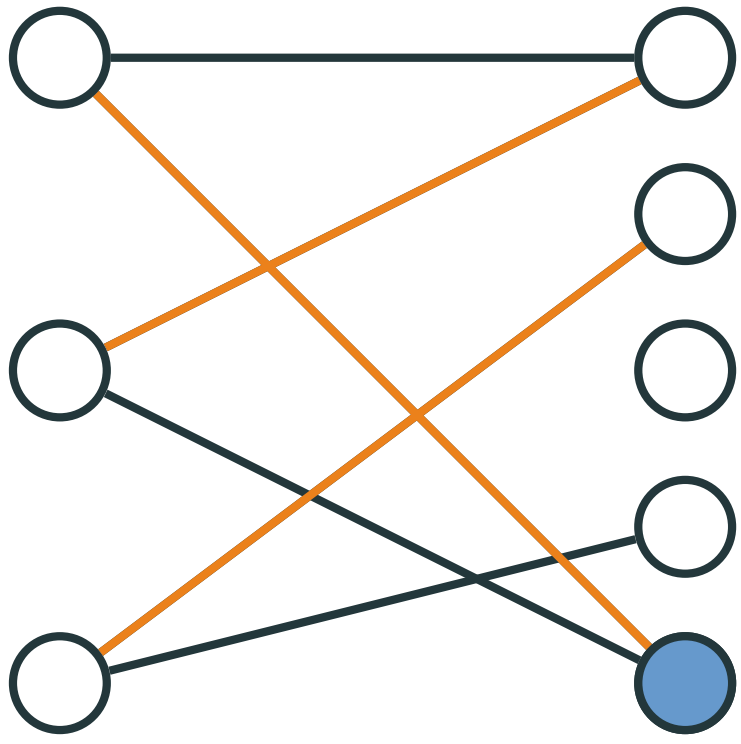
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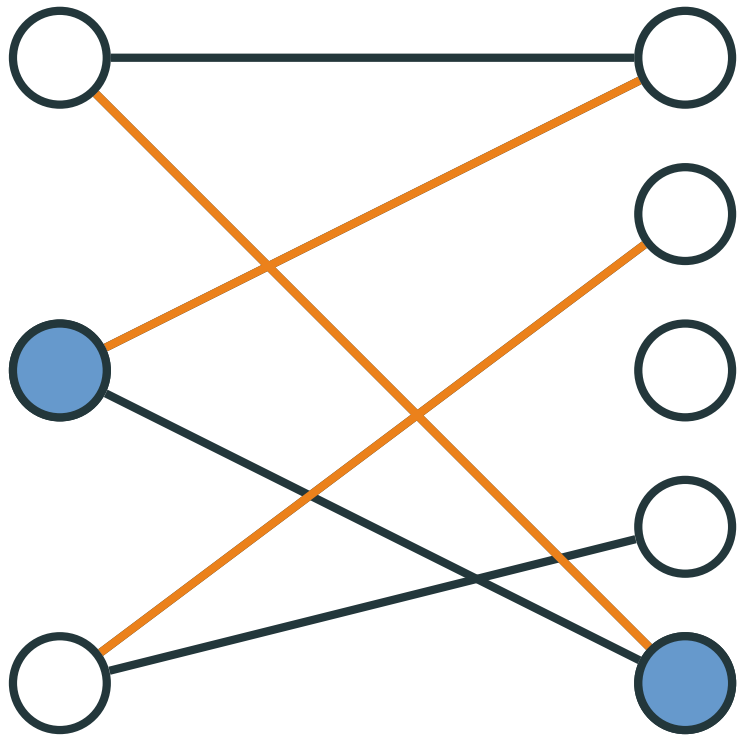
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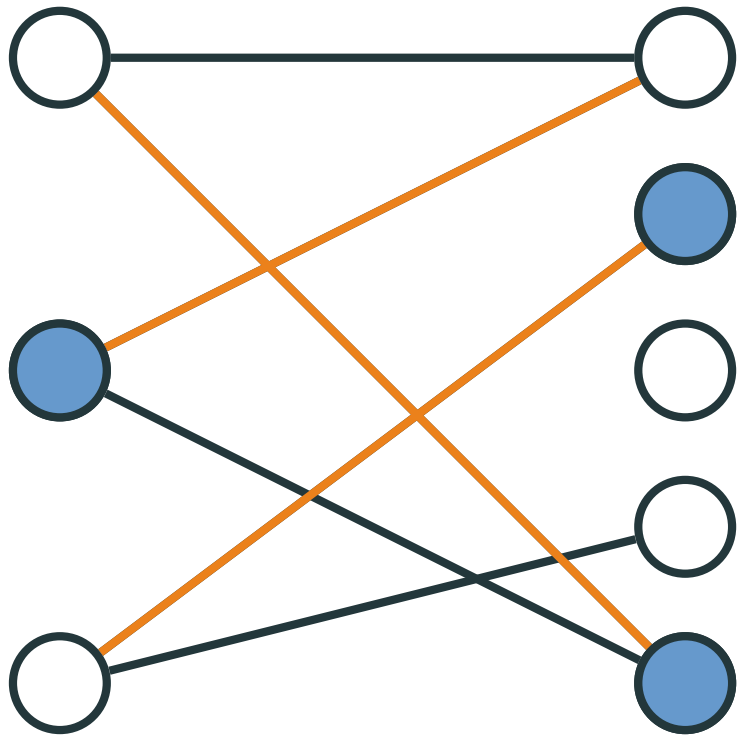
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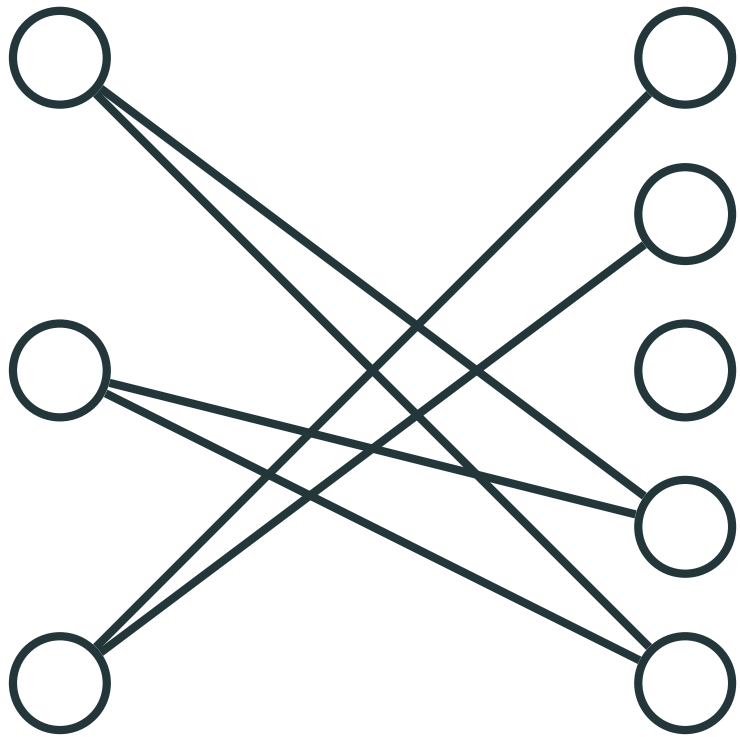
König's Theorem from Hall's Theorem

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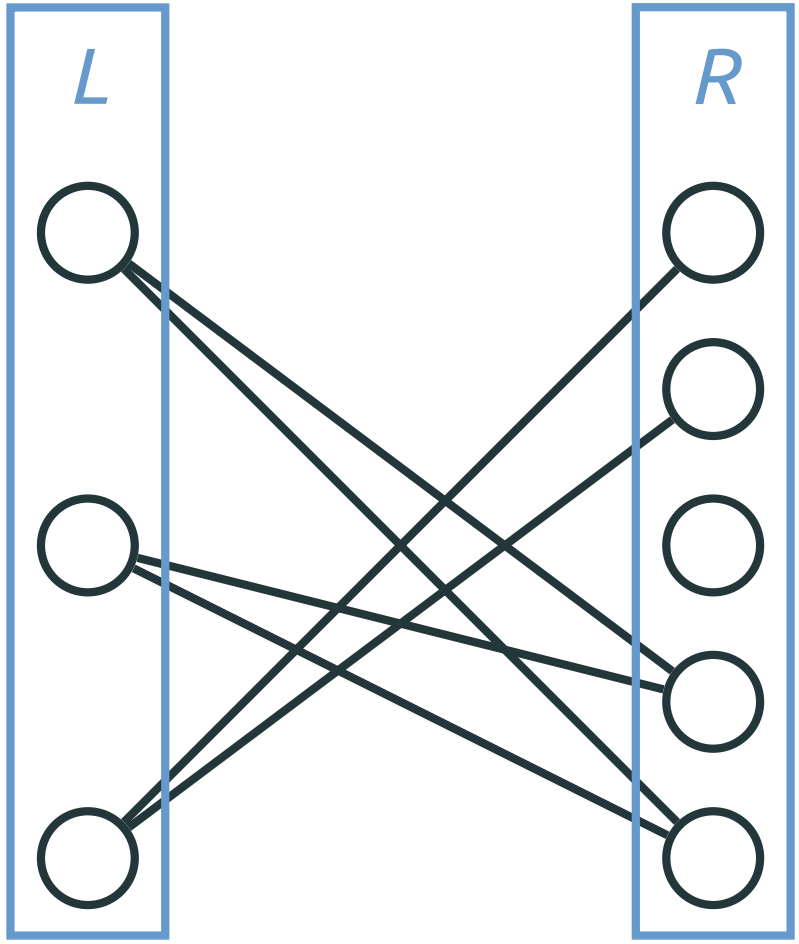
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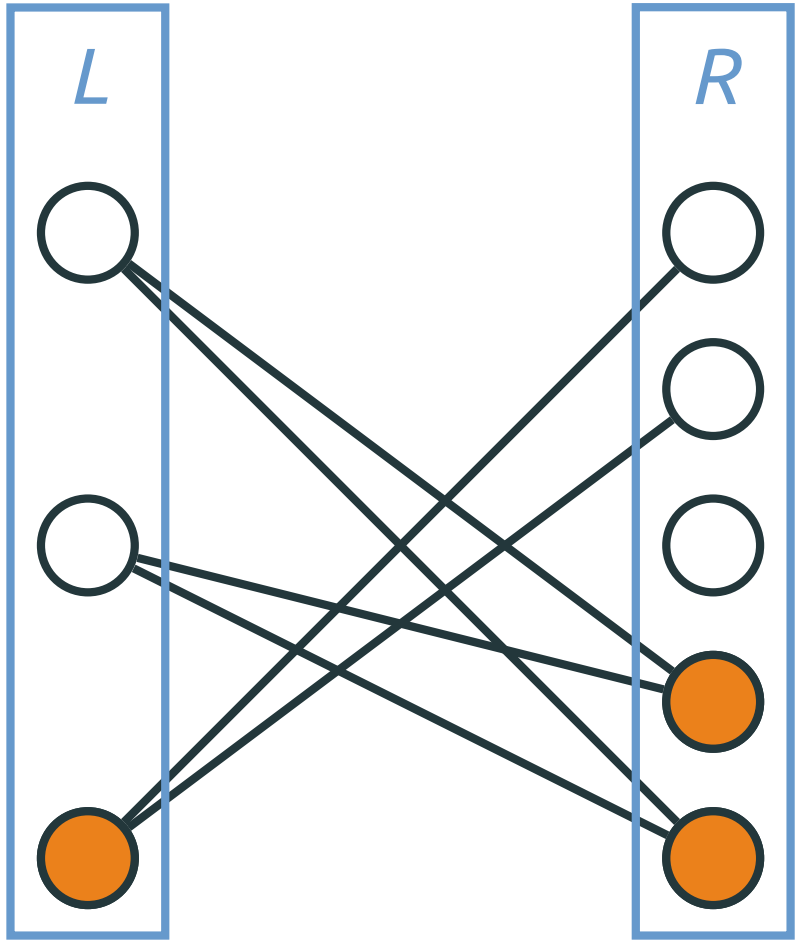


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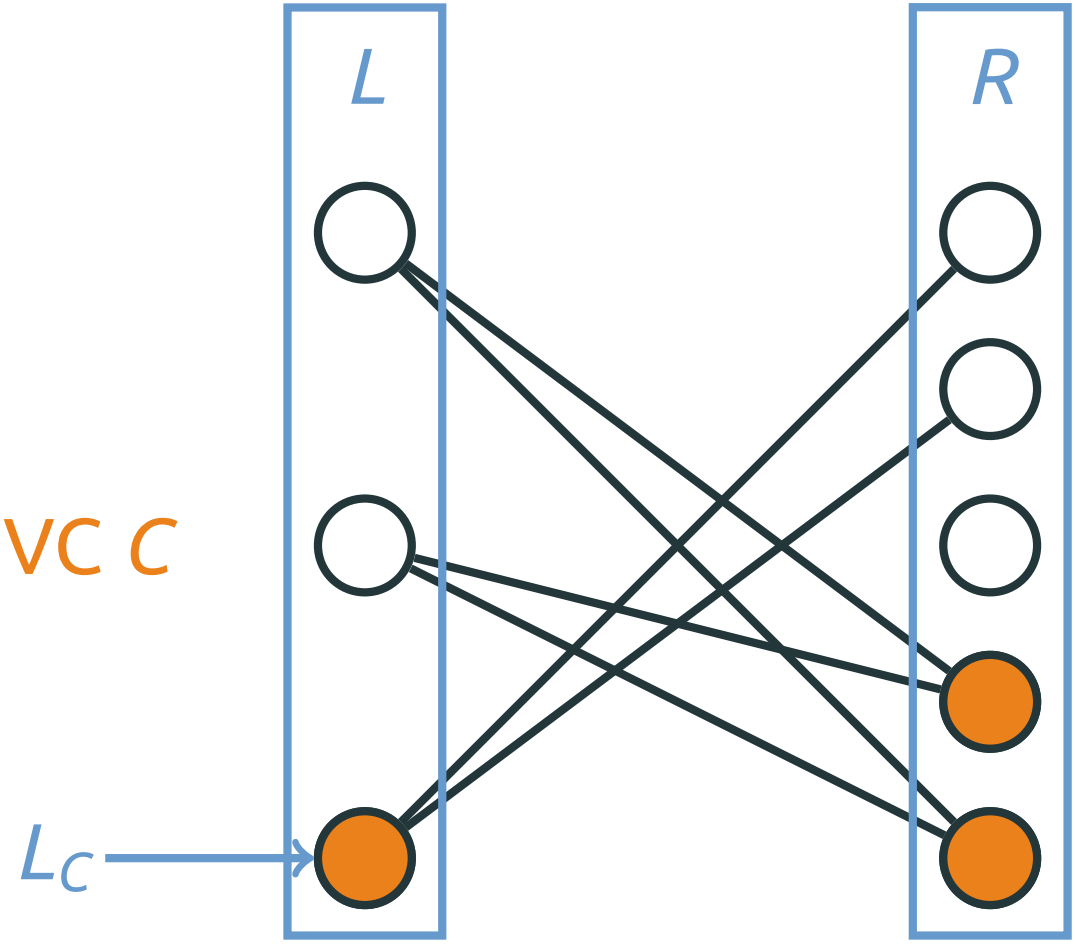
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A Min VC *C*



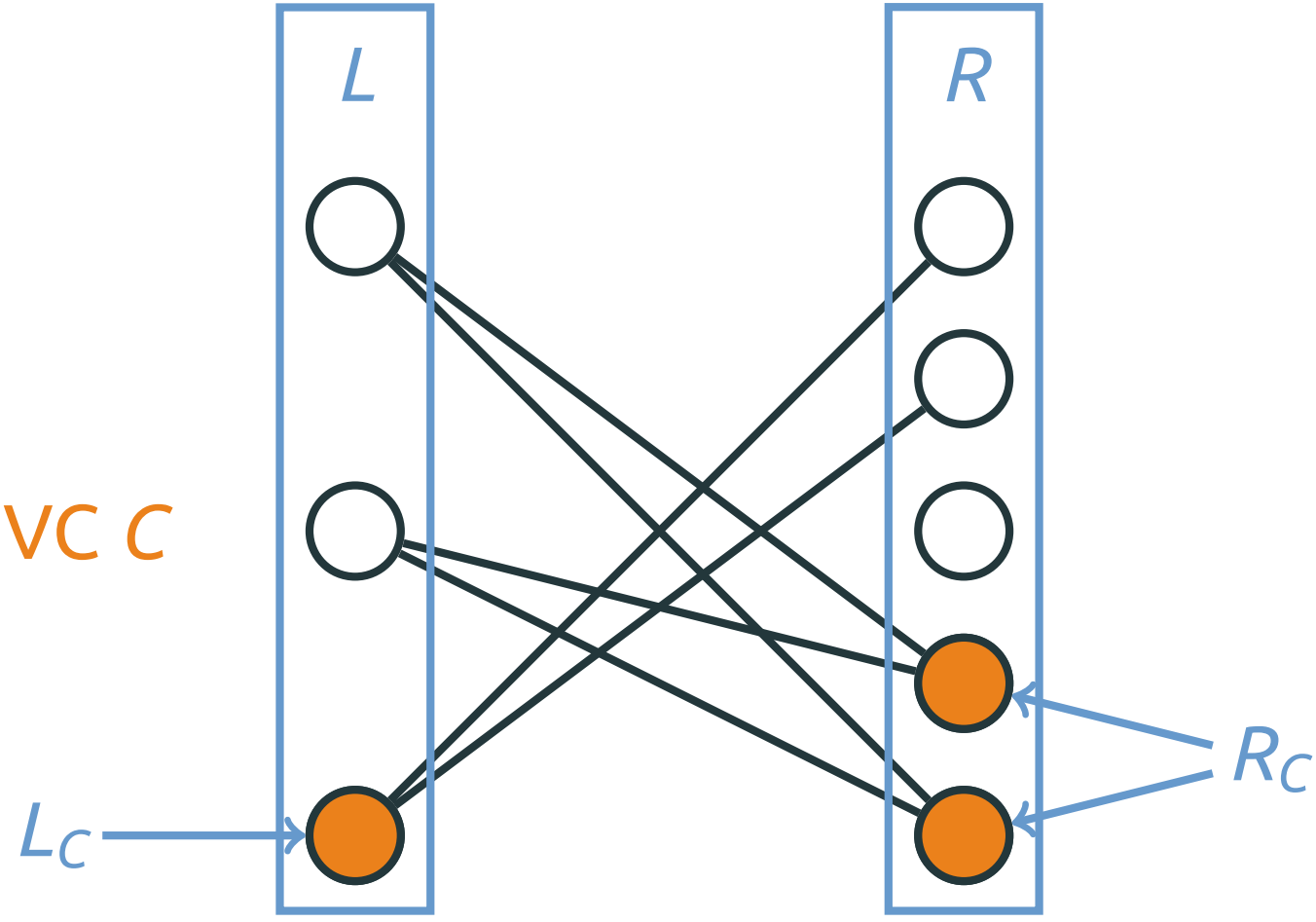
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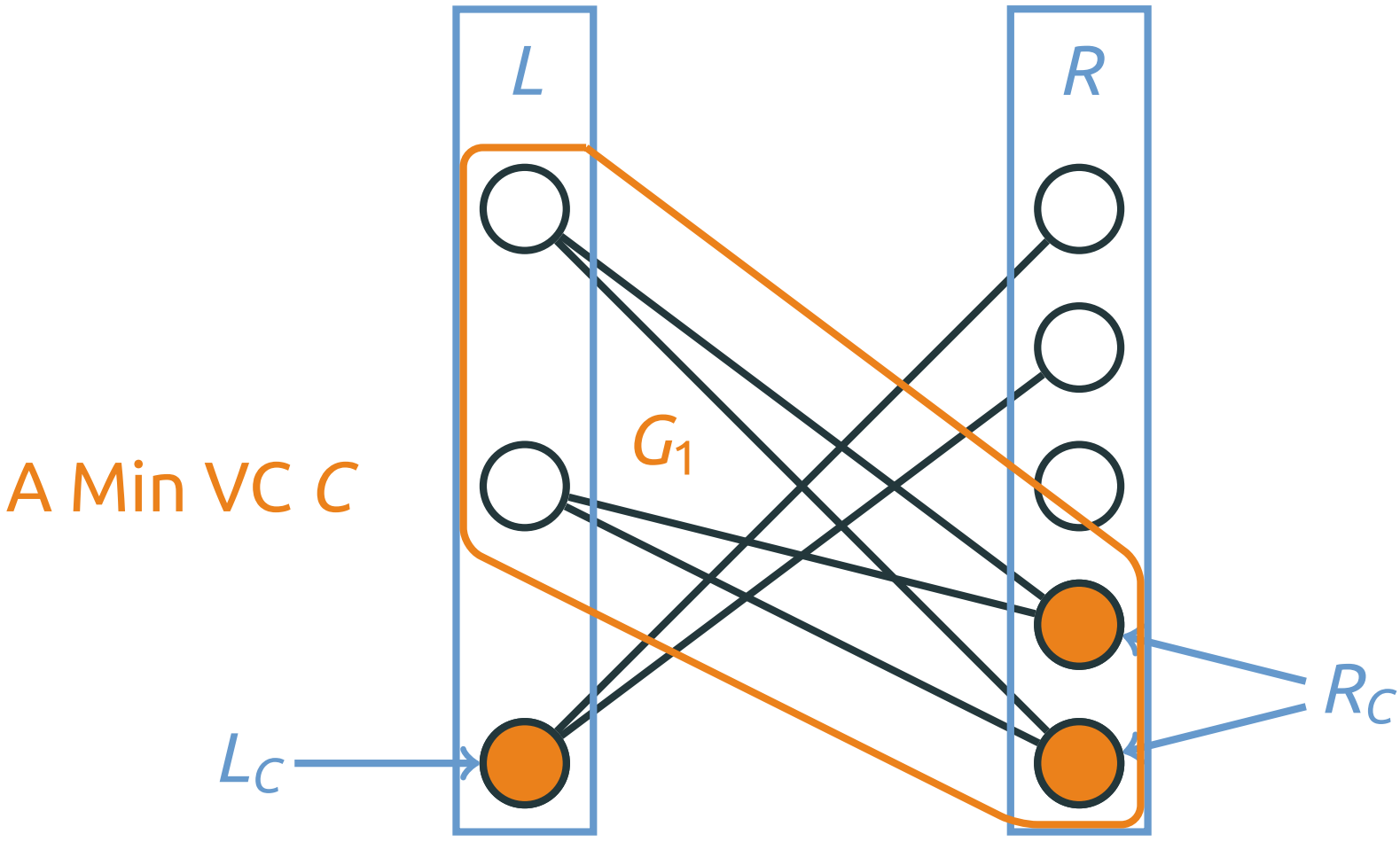


König's Theorem from Hall's Theorem

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König's Theorem from Hall's Theorem



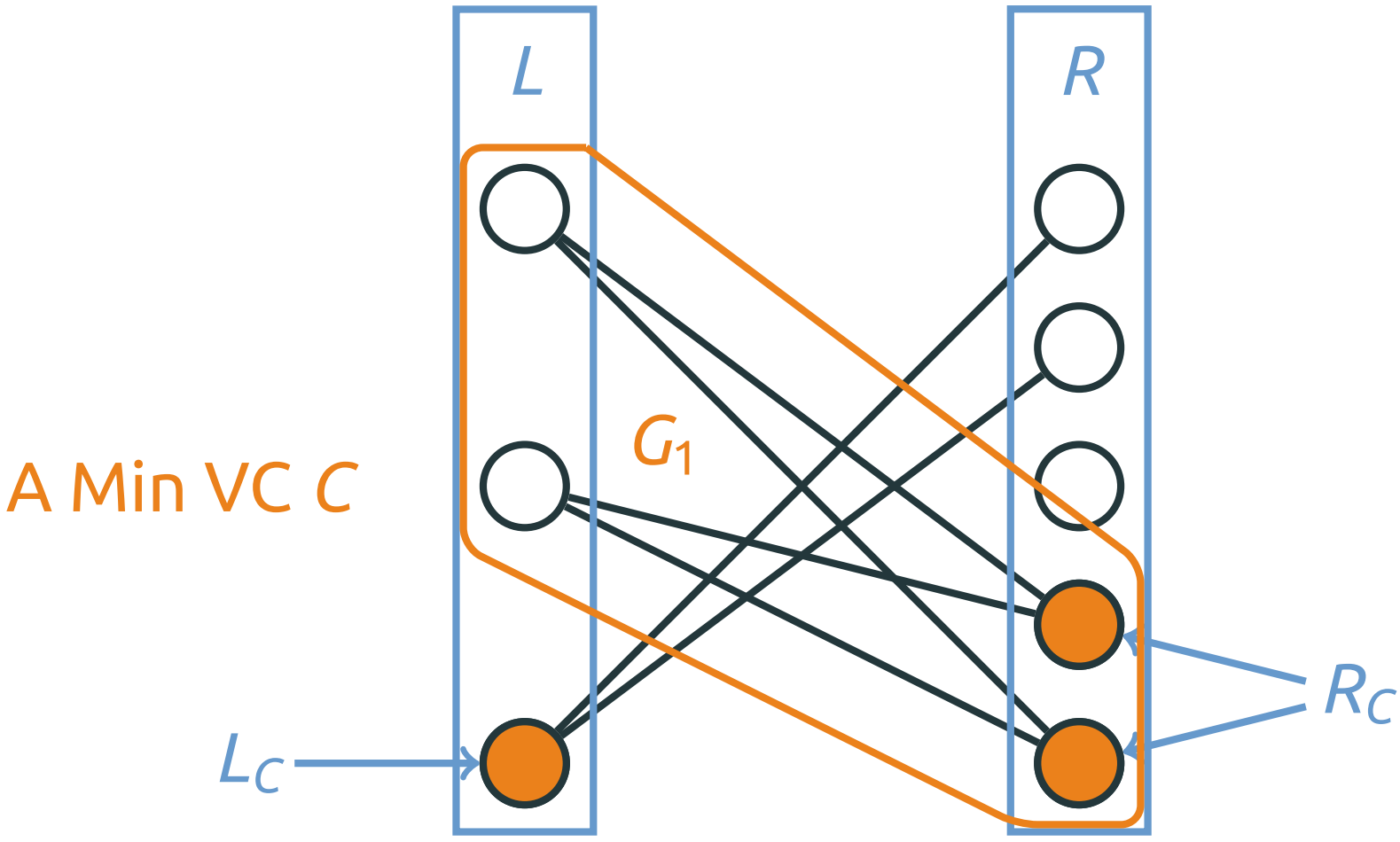
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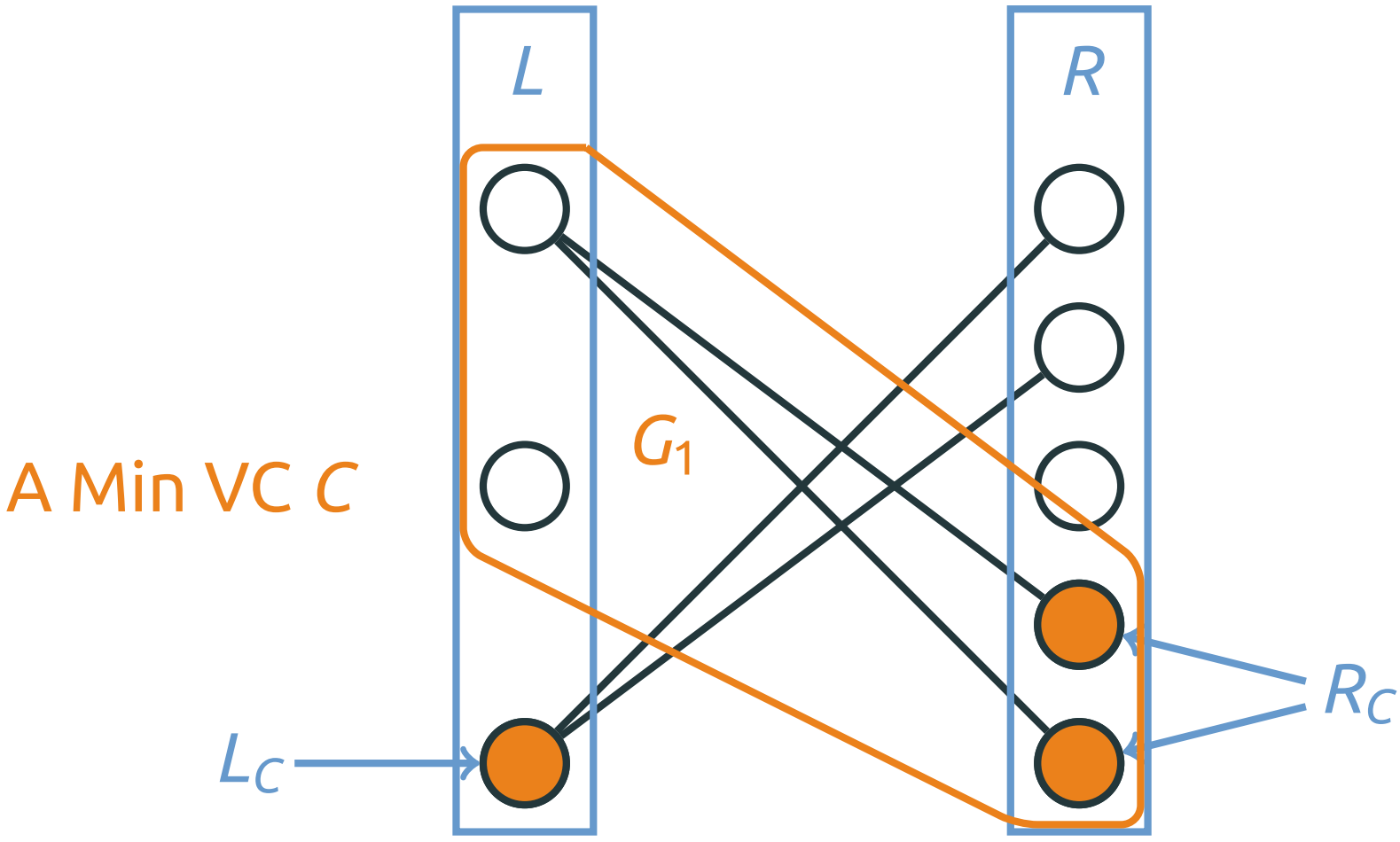
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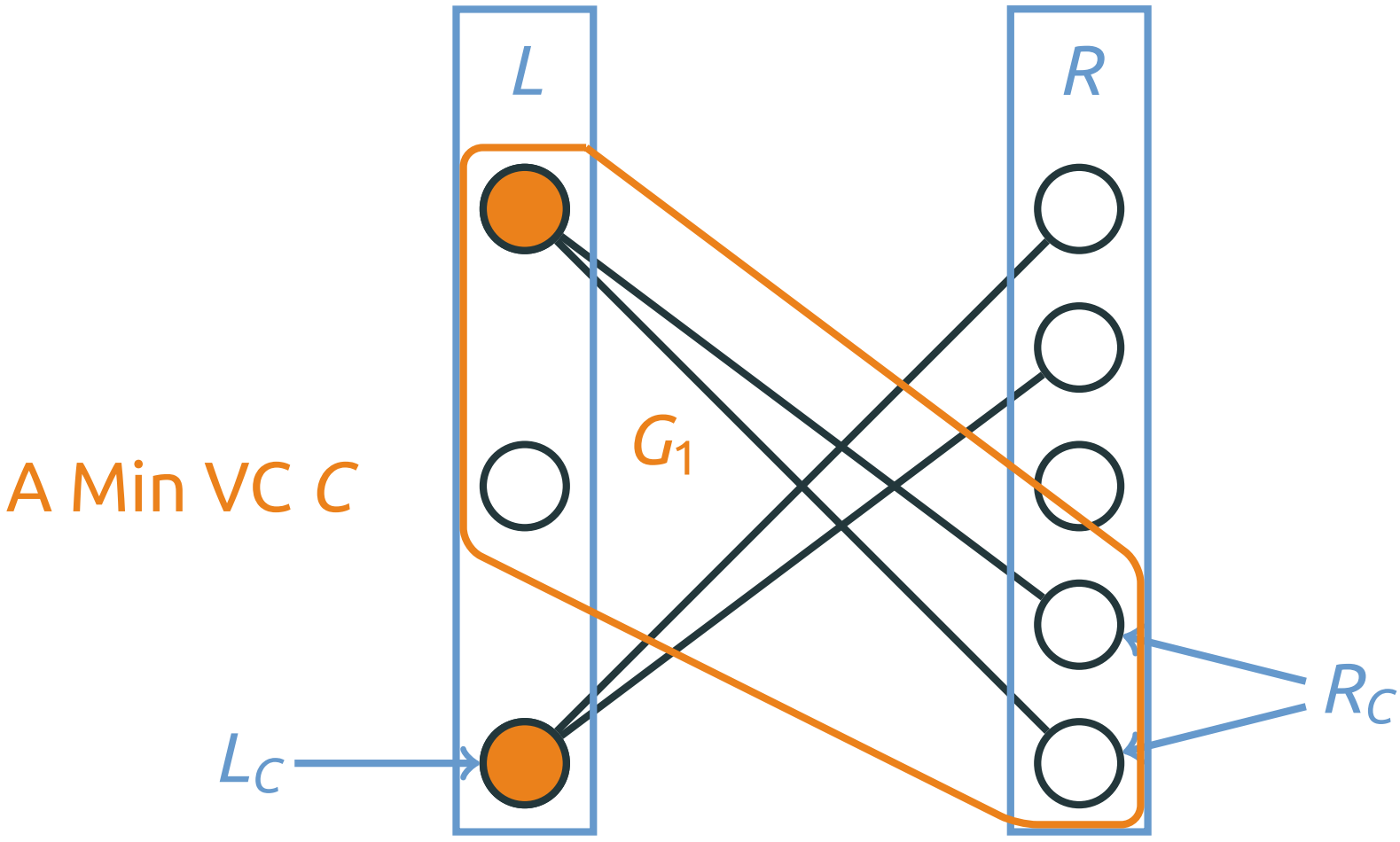
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- Total matching size $|L_C| + |R_C| = |C|$